

Structura Ex Necessitate Standardis Modelis

A relative closure theorem for the Standard-Model bookkeeping interior under fixed load, fixed domain, and certificate-maximal same-domain transport

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Abstract

This paper is *standalone* and *audit-grade*. It does not propose new dynamics or an interpretive layer. It is not a derivation of the Standard Model from first principles, not a uniqueness theorem over all quantum field theories, and not a prohibition on BSM or UV completions. Its target is a relative closure theorem for the Standard-Model bookkeeping interior induced by a declared load, fixed scope, fixed-domain comparison class, and explicit admissibility envelope. Admissibility is not used as a synonym for empirical-output agreement: it also requires stable identity fixation, transport completeness, admissible continuation structure, irreversible reuse of construction records, and exclusion of post-hoc repair.

The paper audits the internal closure of a fixed renormalizable chiral gauge load (Definition 1.2) inside the fixed-domain comparison class (Definition 1.8) under a single governance requirement: *standing* is what survives identity-preserving redescription together with explicit, scope-indexed regime transport. Every collapse is exhibited as a *transport certificate* (Definition 2.57): the skins are named, the transport map is written, the invariant witness set is computed, and witness rank is compared to apparent parameter rank. Low-energy sector results include charged-current matching to G_F , electromagnetic matching to e , Yukawa-mass identity relations, $\bar{\theta}$ as the unique CP-odd QCD invariant, oscillation invariants, and the longitudinal-Goldstone amplitude-class collapse. Discrete-skeleton forcing is then made explicit under the minimal chiral skeleton with one Higgs doublet: anomaly cancellation forces the hypercharge class and guarantees existence of the charged Yukawa operators; hypercharge normalization is shown redundant; tree-level FCNC are excluded.

The central result is typed: within the declared load and the maximal certificate-bearing same-domain admissibility envelope, no further identity-preserving completion exists without introducing new load, leaving the scope, changing the quotient/continuation structure, changing the witness ledger, or invoking a standing-failure mode. The admissibility envelope is not used as an arbitrary filter: any proposed extra same-domain generator must either be a finite standing-preserving composition or factorization already covered by the envelope, or else move an anchor, tensor witness, quotient, or continuation structure. A transformation outside the envelope therefore cannot be both same-domain and standing-preserving unless it supplies a transport-complete certificate that keeps the fixed-domain clauses unchanged; absent that certificate it is not a rival same-scope construction but a typed domain change or collapse. The full renormalizable witness normal form is then stated, including which parameters remain irreducible inputs and why their numerical values are not internally fixed without adding a value-fixation anchor, a new tensor witness, or an inadmissible selector. The fixed-domain comparison class, counterexample burden, anchor/tensor/skin role discipline, repair-exhaustion rule, latent-structure exclusion, envelope maximality, admissibility non-reduction, and no-fifth-locus closure are stated explicitly: any alleged same-scope escape must either supply a witness inside the fixed domain, move an anchor, change tensor witnesses, alter the quotient/continuation structure, or collapse as skin. The sector calculations use standard QFT mechanisms; the added contribution is the admissibility-normalized closure ledger and its counterexample burden.

Contents

1	Scope and non-negotiables	3
1.1	Method posture	3
1.2	Construction-interface bridge and scope lock	3
1.3	Claim typing: relative closure, not external Standard-Model derivation	4
1.4	Relation to standard redundancy, EFT matching, and duality language	4
1.5	Declared load	5
1.6	Declared load is an envelope, not a hidden construction anchor	5
1.7	Fixed-domain comparison class and counterexample burden	6
2	Governance layer: admissibility, transport, witnesses	7
2.1	Minimal coherence forces the admissibility interface	7
2.2	Meta-closure: uniqueness and maximality	10
2.3	Typed primitives	18
2.4	Anchor/tensor/skin role discipline	21
2.5	Standing failures: the diagnostic taxonomy	22
2.6	The engine theorem	22
2.7	Transport certificates: how every collapse will be audited	23
3	Parameter normal form: fully worked construction witnesses and collapses	23
3.1	Charged-current collapse to G_F (fully worked)	23
3.2	Electromagnetic collapse to e (fully worked)	26
3.3	Yukawa–mass identity relation (fully worked, scope-explicit)	31
3.4	QCD CP–odd collapse to $\bar{\theta}$ (fully worked)	32
3.5	Neutrino oscillation invariants (fully worked)	33
3.6	Longitudinal–Goldstone identity transport (worked construction-witness skeleton) . .	34
3.7	Parameter normal form (global statement)	35
4	Forced discrete skeleton: operators, anomalies, charges, FCNC	35
4.1	Operator existence is transport-invariant	35
4.2	Anomaly cancellation and Yukawa compatibility (full algebra)	35
4.2.1	One-generation notation (right-handed hypercharges)	35
4.2.2	Yukawa gauge-invariance conditions	36
4.2.3	Anomaly cancellation conditions (one generation)	36
4.3	Hypercharge normalization and charge quantization	37
4.4	Absence of tree-level FCNC (fully worked)	38
5	Mixing and rephasing: saturation and redundancy (fully shown)	39
5.1	Parameter counting and invariants	39
5.2	Saturation and redundancy statements	40
6	Exhaustion: no further admissible identity-preserving completion remains	40
6.1	What “identity-preserving completion” means here	40
6.2	Admissibility envelope and a normal-form map	40
7	Capstone: the Standard-Model bookkeeping interior is closed (standalone statement)	45

8	Certification: irreducible inputs and structural non-derivability	45
8.1	The full renormalizable scope and the witness ledger	45
8.2	Relational identities vs. internal value fixation	47
9	Hostile-referee attack map (standalone)	48
A	Cubic anomaly check algebra	51
B	Fujikawa construction of the chiral Jacobian and the θ shift	51
B.1	Setup: mode expansion and the measure	51
B.2	Axial rotation and the formal Jacobian	52
B.3	Gauge-covariant regulator and reduction to a local trace	52
B.4	Compute $\not{D}^2$ explicitly	53
B.5	Heat-kernel expansion: isolate the γ_5 term	53
B.6	Coincident-point kernel and the momentum integral	54
B.7	Spinor and color traces	54
B.8	Assemble the coefficient	55
B.9	Read off the θ shift	55
C	Broken-gauge Ward/Slavnov–Taylor identity and longitudinal–Goldstone identity transport	55
C.1	Gauge fixing with an auxiliary field	56
C.2	Ghost term and BRST transformations	56
C.3	F^a is BRST-exact on shell	56
C.4	Vanishing of BRST-exact insertions between physical states	57
C.5	From $\partial \cdot W^a$ insertions to amputated amplitudes	57
C.6	Longitudinal polarization and the high-energy limit	57
C.7	Identity-transport theorem (tree level, audit form)	58

1 Scope and non-negotiables

1.1 Method posture

This paper is eliminative. It introduces:

- no new primitives, no new interpretive layer, and no new anchoring assertions;
- no compensating fields, auxiliary sectors, additional $U(1)$ factors, or exotic charge lattices;
- no probabilistic/typicality repairs, no external standing import, and no privileged representation.

All results are obtained by explicit calculation plus one governance rule: *standing is what survives admissible redescription and admissible transport*. Anything else is redundancy or standing failure.

1.2 Construction-interface bridge and scope lock

This framework does not treat inferential activity as a primitive process. It enforces the structural conditions under which statement-bearing constructions can persist without collapse. In external bridge language only, what is commonly called “reasoning” is reinterpreted here as persistence of admissible construction trajectories under identity-preserving constraints.

Non-optional use. Any construction system capable of irreversible, identity-preserving evaluation must instantiate these constraints or fail to define its own construction space. A system that permits post-hoc reindexing of committed construction records does not provide an alternative same-domain identity practice; it loses the standing carrier required for comparison.

No interpretive layer. This framework does not introduce an interpretive layer. It characterizes necessary structural conditions on admissible construction systems. Any system failing to satisfy these conditions cannot sustain statement-bearing states.

Scope lock. All statements are made within a fixed domain. Cross-domain transport without admissible identity preservation is undefined.

1.3 Claim typing: relative closure, not external Standard-Model derivation

The central assertion is deliberately typed. The paper does not assert that the Standard Model is the only logically possible quantum field theory, nor that the gauge group, representation skeleton, generation count, or numerical values of the remaining witnesses are forced from no prior load. The assertion is narrower and stronger within its domain: once the declared renormalizable chiral gauge load, fixed scope, admissible transport interfaces, and witness discipline are fixed, no further same-scope standing-bearing completion is available. Any alleged counterexample must therefore remain inside the same fixed-domain comparison class (Definition 1.8); otherwise it is a different load, a different scope, or a different construction problem.

Remark 1.1 (Why fixed-load closure is non-vacuous). The fixed load does not by itself sort the bookkeeping into standing content and surplus. Even after the load is fixed, there remain apparent freedoms: gauge choice, flavor basis, rephasing, hypercharge normalization, mediator versus EFT parameterization, low-energy Wilson coefficients, CP-odd phase splits, longitudinal versus Goldstone presentations, renormalization-scheme carriers, and possible selectors or repair rules. The paper’s nontrivial task is to show which of these are anchors, which are tensor witnesses, and which are skin. That sorting requires explicit transport certificates and quotient factorization; it is not supplied by the load alone.

1.4 Relation to standard redundancy, EFT matching, and duality language

The local ingredients used below—gauge fixing, field redefinition, flavor-basis rotation, anomaly cancellation, Yukawa diagonalization, field-redefinition invariance, EFT matching, and decoupling—are standard tools in quantum field theory [1–8]. The novelty asserted here is not that any one local calculation is new. The added structure is the fixed-domain closure ledger: every proposed same-scope distinction must be assigned to anchor, tensor witness, or skin; every transport must preserve identity across the declared regime interface; and every alleged completion must either supply a new invariant witness inside the same comparison class or collapse as redundancy/standing failure. Thus the target is not a replacement for standard equivalence, gauge-redundancy, duality, algebraic-QFT, or EFT practice [15–20], but a fail-closed audit discipline for when those practices are allowed to carry standing. Standard empirical-equivalence language asks whether two presentations agree on outputs; Definition 2.5 makes explicit that the present discipline additionally asks whether that agreement survives transport, continuation, quotient preservation, and irreversible reuse.

1.5 Declared load

We fix the following *declared load* for the entire paper.

Definition 1.2 (Declared load). *The declared load $\mathcal{L}$ consists of:*

- (i) *the renormalizable $SU(3)_c \times SU(2)_L \times U(1)_Y$ gauge framework;*
- (ii) *minimal chiral fermion content in the usual representation skeleton (three generations allowed but not required for the algebraic construction sequences);*
- (iii) *a single electroweak Higgs doublet H (with conjugate $\tilde{H} \equiv i\sigma^2 H^*$ available as usual);*
- (iv) *no additional Higgs multiplets, no vectorlike patch pairs, no extra abelian factors, and no ad hoc patch operators (non-gauge-invariant or higher-dimensional) introduced as independent primitives; higher-dimensional operators may appear only as transport images (EFT skins) of the renormalizable load.*

Remark 1.3 (What is and is not internally constructed). Definition 1.2 fixes the renormalizable chiral gauge skeleton whose internal closure is audited. This paper does *not* attempt to derive the gauge group or the representation skeleton from scratch; doing so would be a different project with a different scope. The present construction sequences concern (i) collapse of *apparent* parameter freedoms under scope-indexed transport, and (ii) closure of *same-scope* modifications relative to the fixed load.

Remark 1.4 (Fail-closed discipline). *Inadmissible* is used in the fail-closed sense: if a purported move cannot be made standing-preserving under admissible identity-preserving redescription and explicit transport inside $\mathcal{L}$, then it is rejected as either (a) redundancy (no standing), or (b) standing failure (requiring hidden load, representational privilege, illicit scope transport, or compensating structure).

1.6 Declared load is an envelope, not a hidden construction anchor

A frequent misdiagnosis in “uniqueness” analyses is to treat a fixed load choice as a covert assumption smuggled in to force the desired answer. Here we make the opposite move: we *type* the assertion. The paper does *not* assert that no other loads are logically possible. It asserts that, *relative to the scope and declared load*, there is a unique *minimal standing-preserving interior*, and that purported “extensions” asserting same-scope status in the same scope are either redundant or scope-changing.

Definition 1.5 (Scope-preserving vs. scope-changing modification). *Fix a scope S (Definition 2.34). A proposed modification Δ of the bookkeeping is:*

- (i) *scope-preserving if it reproduces exactly the same invariant witness set for S (up to admissible identity-preserving redescription) without introducing additional standing-relevant witnesses;*
- (ii) *scope-changing otherwise (it enlarges/changes the witness set, the admissibility envelope, or the regime content).*

Proposition 1.6 (Added-load dichotomy). *Let S be a fixed scope and $\mathcal{L}$ the declared load. Consider adding extra fields/operators/sectors $\Delta\mathcal{L}$. If $\Delta\mathcal{L}$ introduces no new invariant witnesses for S , then $\Delta\mathcal{L}$ is redundancy for S (it is removable by admissible identity-preserving redescription/transport, e.g. integrating out). If $\Delta\mathcal{L}$ does introduce new invariant witnesses for S , then it is a scope change and therefore not a nontrivial scope-preserving extension of the SM interior.*

Construction witness. If $\Delta\mathcal{L}$ introduces no new invariant witnesses for S , then by Definition 2.49 it contributes no standing-relevant data for S and is removable without changing witness content. If $\Delta\mathcal{L}$ introduces new invariant witnesses, then by Definition 1.5 the scope has changed: the modified bookkeeping describes strictly more (or different) standing content. In neither case does $\Delta\mathcal{L}$ produce a genuine same-scope enlargement of standing content. $\square$

Remark 1.7 (What is *not* being asserted). Proposition 1.6 is not a “BSM ban.” It is a typing statement: *inside* the present scope and declared load, additional structure is either redundant or a scope change. Scope changes may be physically motivated; they are simply different projects and are not adjudicated here.

1.7 Fixed-domain comparison class and counterexample burden

The closure assertions below are not global assertions over every mathematically imaginable theory. They are fixed-domain assertions over the same-scope comparison class in which the Standard-Model bookkeeping interior is evaluated.

Definition 1.8 (Fixed-domain / same-scope comparison class for this paper). *A proposed modification, redescription, selector, transport, parameter split, value-fixation rule, or completion remains same-scope for a fixed scope S iff all of the following are preserved simultaneously:*

- (i) *the declared load $\mathcal{L}$ and the discrete representation skeleton admitted for S ;*
- (ii) *the regime, admitted tests, and error envelope specified by S ;*
- (iii) *the standing relation $\sim_S$ and hence the standing quotient $P(\mathcal{S})/\sim_S$;*
- (iv) *the invariant witness family for S , up to identity-preserving reparameterization;*
- (v) *the licensed admissible transports, histories, and composition structure used to compare skins;*
and
- (vi) *the role typing of components as anchors, tensor witnesses, or skin (Section 2.4).*

A move altering any item in (i)–(vi) is not a same-scope refinement. It is a scope change, new load, or a standing failure, depending on which item is moved.

Remark 1.9 (The comparison class is not an ad hoc shield). Definition 1.8 is the minimal domain in which the paper’s target phenomenon is well-typed: standing-bearing evaluation, identity-preserving redescription, explicit regime transport, and irreversible reuse of construction records. Rejecting this comparison class does not produce a same-scope counterexample; it removes the fixed-domain object being audited.

Remark 1.10 (Out-of-class movement is not an internal objection). A transformation, formalism, or operator outside the fixed-domain comparison class can still be studied as a larger-domain construction. It cannot function as a same-scope counterexample unless it preserves standing, transport completeness, and fixed-domain comparability for the original S . This is a necessity condition, not a terminological convention: without those three carriers, there is no identity-preserving map on the standing quotient whose failure could refute the closure result.

Theorem 1.11 (Admissibility constraint on same-domain counterexamples). *A genuine counterexample must exhibit a same-scope standing-bearing difference while preserving all six conditions of Definition 1.8. Equivalently, it must show a new standing-relevant operator, parameter, or witness*

that changes no load boundary, no admitted test family, no error envelope, no quotient relation, no transport/history structure, and no anchor/tensor/skin role assignment. Absent that construction witness, the proposal is bookkeeping, scope change, load augmentation, or representational privilege.

Construction witness. If a proposal changes any one of the six fixed-domain clauses, it has moved the domain and cannot be a same-scope counterexample. If it preserves all six clauses while introducing no new witness, then it changes no standing class and factors through the normal form. If it preserves the clauses and introduces a new witness, then the witness must be exhibited inside the same quotient/transport structure; otherwise the asserted difference is unwitnessed load or representational privilege. These alternatives exhaust the same-domain possibilities. $\square$

Remark 1.12 (Theorem typing convention and global-reading exclusion). Unless explicitly stated otherwise, all closure, uniqueness, exhaustion, and forcing statements in this paper are relative to the fixed-domain comparison class of Definition 1.8. In Sections 6–7 the additional reference class is the admissibility envelope $\text{Ad}_{\mathcal{L}}$ (Definition 6.1). A proposal that changes the load, admitted tests, error discipline, transport rules, quotient relation, or witness family may be an interesting larger-domain construction, but it is not a counterexample to a same-scope closure theorem.

2 Governance layer: admissibility, transport, witnesses

2.1 Minimal coherence forces the admissibility interface

The governance primitives in this section are not decorative. They are¹ forced by minimal coherence requirements for any scope-indexed bookkeeping practice.

Definition 2.1 (Minimal coherence requirements). *Fix a scope S . A construction bookkeeping practice for S is minimally coherent if it satisfies:*

- (MC1) **Stable identity fixation.** *Tracked objects in S must remain fixed across lawful identity-preserving transformations; the identity of “the same” constructed object may not depend on a privileged internal parameterization.*
- (MC2) **Lawful identity-preserving transformation.** *Multiple internal presentations may be used (field variables, bases, gauge-fixings, parameterizations), and such transformations are permitted so long as they preserve S -outputs.*
- (MC3) **Irreversible commitment.** *Once a construction record is fixed, it cannot be retroactively reindexed by choosing a different internal presentation; standing must be maintained under irreversible reuse of records.*
- (MC4) **Non-triviality.** *The bookkeeping has at least one witness-extraction output that distinguishes some admissible cases inside S (otherwise all constructed states collapse vacuously).*

Remark 2.2 (MC1 is a standing postulate, not an ontic prohibition). The minimal-coherence axioms are governance constraints for *standing* in a fixed scope S . In particular, **MC1** says that within the same-scope interior we do not treat distinctions with no admissible S -witness as identity-bearing. This is *not* the metaphysical assertion that nature contains no additional structure; it is the bookkeeping discipline that such additional structure, if asserted, must appear as added load or enlarged scope and hence be typed as a scope change (Proposition 1.6).

¹In the same sense that Myhill–Nerode equivalence is not “chosen” but forced once one demands a canonical presentation of future-distinguishability classes.

Remark 2.3 (Minimal coherence is a domain contract, not hidden dynamics). The minimal-coherence clauses do not smuggle in the Standard Model or any numerical physics. They specify the minimum conditions under which a fixed-domain construction can reuse identity-bearing records across lawful presentation changes. A framework that rejects stable identity fixation, lawful identity-preserving transformation, irreversible commitment, or nontrivial witness extraction is not a rival same-scope bookkeeping for S ; it is a different construction problem and must be typed outside Definition 1.8.

Remark 2.4 (Irreversible commitment is not a dispensable bookkeeping preference). The irreversible-commitment clause is used only after a construction record is allowed to bear on identity, transport, or subsequent witness extraction. Before such use, a presentation can remain provisional. Once used, retroactive representative selection, silent relabeling, or generate-then-filter admission would change the identity carrier after the fact. That is precisely the AASC no-deferred/no-post-selection failure: later auditing or filtering may define a new construction lineage, but it cannot retroactively make the earlier committed lineage admissible.

Definition 2.5 (Output agreement vs. admissibility). *For a fixed test family in a fixed regime, output agreement means equality of the admitted witness outputs within the declared error discipline. Admissibility is stronger: it requires output agreement together with stable identity fixation, transport completeness, preservation under admissible continuations, irreversible reuse of construction records, and absence of same-lineage repair. Thus two presentations may agree on a snapshot family of outputs while still failing to be admissibly identical if one loses standing-relevant data under transport or continuation.*

Proposition 2.6 (Output agreement alone is insufficient for standing). *Within the fixed-domain comparison class, a construction sequence that uses output agreement alone as a same-scope standing criterion collapses unless it also supplies admissible identity-preserving transport and quotient preservation in the sense of Definition 1.8.*

Construction witness. Output agreement can hold accidentally, locally, or after forgetting data. A truncated EFT without its error envelope, a gauge-fixed presentation treated as privileged, and a dual presentation lacking quotient-injective transport can all reproduce a selected list of outputs while failing to preserve standing under continuation or regime change. The missing datum is not another output; it is the identity-preserving carrier that keeps the same constructed object fixed across redescription, update, and irreversible reuse. $\square$

Lemma 2.7 (Non-reducibility of admissibility to snapshot output agreement). *Within the fixed-domain comparison class, admissibility is not reducible to snapshot output agreement except in the degenerate case where the admitted output family already contains every admissible continuation, every transport-complete witness, and every irreversible reuse condition. More precisely, if there exist parameter points $\mathbf{p}, \mathbf{p}'$ and an admissible continuation u such that $\mathbf{p}$ and $\mathbf{p}'$ agree on the primitive output family but some admitted output distinguishes $u(\mathbf{p})$ from $u(\mathbf{p}')$, then any construction identifying $\mathbf{p}$ and $\mathbf{p}'$ by snapshot output agreement alone collapses under irreversible reuse. Likewise, if two skins agree on a selected output list but no transport-complete quotient map preserves the standing witnesses, output agreement is insufficient for admissibility.*

Construction witness. Snapshot output agreement checks only a current test face of the construction. Admissibility additionally requires stability under admissible continuations, transport-complete comparison of skins, quotient preservation, and irreversible record reuse. If $u(\mathbf{p})$ and $u(\mathbf{p}')$ become distinguishable after a licensed continuation, then the map $[\mathbf{p}] \mapsto [u(\mathbf{p})]$ is not well-defined on the snapshot quotient: the result depends on the chosen representative. If a skin-to-skin map

loses a witness needed after a regime interface, then the same failure appears as loss of transport completeness. Neither dependence can be repaired after a construction record has been fixed, by MC3. Lemma 2.20 gives the full kernel construction; the present lemma records the immediate non-reducibility consequence. $\square$

Remark 2.8 (Admissibility is not empirical-output equivalence). This distinction blocks the common compression of the paper to ordinary empirical equivalence. Snapshot agreement is only a necessary condition for same-scope standing. A skin that reproduces a finite list of outputs but lacks a transport-complete map, changes the continuation structure, requires a privileged representative, or repairs identity only after a record has been fixed is not an admissible same-scope replacement. The extra burden is precisely what is enforced by Lemma 2.7, Lemma 2.20, Theorem 2.28, and Theorem 2.30.

Remark 2.9 (Axiom denial is a domain change, not an internal counterexample). One may deny stable identity fixation, lawful identity-preserving transformation, irreversible commitment, or non-trivial witness extraction. That denial is permitted as an external bridge move, but it changes the construction system under audit. Inside the fixed-domain comparison class, a counterexample must preserve the minimal-coherence interface and then exhibit a same-domain standing witness that survives quotient reduction.

Remark 2.10 (No post-hoc reindexing of committed construction records). A histories, path-integral, algebraic, or dual presentation may reorganize how records are generated. It cannot, while remaining a same-domain bookkeeping for S , allow a record already fixed as one standing class to be retroactively assigned to another by changing representatives. That move destroys the well-defined action of admissible continuations on the quotient and collapses the identity of the reused record. Thus irreversibility is not an optional psychological or procedural assumption; it is the condition under which stored construction records can be reused without changing the object being tracked.

Remark 2.11 (Standing is a same-domain identity criterion). Standing is not a global prohibition on hidden structure or broader physics. It is the identity criterion for the fixed construction domain. Additional structure becomes relevant to this paper only by entering as new load, a new admitted scope, a new invariant witness, or a new licensed transport. If it enters none of these, it is skin relative to the fixed domain.

Remark 2.12 (Latent-but-non-witnessed structure has no same-scope standing). A latent variable, metadata layer, hidden label, or broader-theory parameter that is not representable by an invariant witness in the declared scope, and that does not alter the standing quotient or admissible transport/history structure, defines no distinct same-scope standing state. If such structure becomes visible only after enlarging the admitted tests, changing the transport interface, or adding new load, then the construction has left the fixed-domain comparison class. This remains so even for an arbitrarily small scope extension: witnessability obtained by changing the admitted tests is a typed domain change, not a same-scope completion. Thus latent existence outside the witness ledger is not a same-scope extension; it is either skin for the present domain or a typed larger-domain construction. If it is retained as identity-bearing while remaining unwitnessed inside the declared domain, it is hidden unwitnessed structure and therefore a standing-failure mode under Definition 2.53.

Theorem 2.13 (Admissibility constraint on the standing quotient and admissibility groupoid). *Let constructed states exist inside a fixed scope S , the declared load $\mathcal{L}$, and the fixed-domain comparison class of Definition 1.8, satisfying minimal coherence (Definition 2.1). Then any construction sequence that treats witness-indistinguishable parameter points as different standing states collapses by standing failure. Conversely, all admissible same-scope constructions satisfy:*

- (i) *The only identity-bearing objects for S are standing equivalence classes $[\mathbf{p}] \in P(\mathcal{S})/\sim_S$ (Definitions 2.46–2.47). In particular, any attempt to treat two parameter points $\mathbf{p} \sim_S \mathbf{p}'$ as distinct standing states introduces representational privilege or hidden unwitnessed structure (Definition 2.53).*
- (ii) *Any assertion that two skins $\mathcal{S}_A, \mathcal{S}_B$ represent the same scope content requires transports that are transport-complete for S (Definition 2.36); equivalently, the induced maps on standing quotients must be injective, and for mutual identity preservation bijective.*
- (iii) *The class of skins equipped with transport-complete admissible transports that admit inverses on standing quotients forms a groupoid $\mathbf{Ad}_S$ (objects: skins for S ; morphisms: transports inducing bijections on standing quotients). Standing is functorial: every morphism in $\mathbf{Ad}_S$ induces a bijection of standing state spaces.*

Construction witness. (i) Let $\mathcal{S}$ be any skin used for scope S . By (MC2), lawful redescrptions are permitted provided they preserve S -outputs. Therefore any internal parameter value that does not change S -outputs cannot be privileged as identity-grounding without violating (MC1): doing so would make tracked-object identity depend on an arbitrary representational choice.

Formally, define $\mathbf{p} \sim_S \mathbf{p}'$ exactly as in Definition 2.46: equality of all S -outputs within the declared error discipline. If $\mathbf{p} \sim_S \mathbf{p}'$ but one insists they are distinct standing states, then the distinction has *no* S -witness. Maintaining the distinction therefore requires either a privileged parameterization (violating MC1/MC2), or extra unwitnessed structure imported to “remember” which point one is in (hidden load), which is precisely a standing-failure mode (Definition 2.53). Thus coherence forces identification of standing states with equivalence classes $[\mathbf{p}]$.

(ii) Consider two skins $\mathcal{S}_A, \mathcal{S}_B$ both purported to describe scope S . If there is no transport-complete map, then there exist $\mathbf{p}, \mathbf{p}'$ in one skin that are S -distinguishable but whose images in the other skin cannot be distinguished by any invariant witness there. Then the identity of the S -tracked object depends on which skin is used, contradicting (MC1) and (MC3). Therefore any coherence assertion of “same scope content” requires transport completeness. Injectivity/bijection on standing quotients is exactly this requirement in quotient form.

(iii) Define $\mathbf{Ad}_S$ to have as objects those skins used for S and as morphisms the admissible transports that induce bijections on standing quotients (equivalently, are transport-complete and admit inverses up to $\sim_S$). Composition of such morphisms is again transport-complete and induces bijections on quotients; identity morphisms are given by trivial transports. Existence of inverses on quotients makes this a groupoid. Functoriality of standing is then immediate: each morphism induces a bijection on $P(\mathcal{S})/\sim_S$ by construction; Lemma 2.48 provides the explicit statement. $\square$

2.2 Meta-closure: uniqueness and maximality

Referee Construction-Witness Obligations. To refute the meta-closure results of Section 2.2, a critic must do *at least one* of the following: (i) explicitly deny at least one Minimal Coherence axiom (MC1–MC4), stating which condition of stable identity fixation, irreversibility, lawful redescription, or non-trivial construction record is rejected; (ii) exhibit a concrete standing presentation (X, r) satisfying (MC1–MC4) that reproduces all S -outputs yet is not isomorphic (via the uniquely induced bijection) to the standing quotient $P(\text{Skin})/\sim_S$; or (iii) produce a same-scope morphism that introduces a new invariant witness without violating transport completeness, minimal coherence, or declared load. Absent one of (i)–(iii), objections reduce to scope change, load augmentation, or representational privilege, each already classified.

The previous subsection established that minimal coherence forces the quotient notion of standing and the groupoid form of same-scope identity transport. A hostile reader may still attempt a meta-level escape: “perhaps there exists some *other* equally coherent identity/transport scheme.” Here we close that route in the same way as a Myhill–Nerode construction sequence closes “alternative” state presentations: the standing quotient is not one option among many; it is the unique standing state space compatible with scope-indexed witness output invariance and stable identity fixation.

Definition 2.14 (Standing presentation for a scope). *Fix a scope S and a skin $\mathcal{S}$ with parameter space $P(\mathcal{S})$. A standing presentation for S is a triple (X, r, Out_X) where:*

- (i) X is a set intended to serve as the standing state space for S ;
- (ii) $r : P(\mathcal{S}) \rightarrow X$ is a surjection (“representation map”);
- (iii) for every $\mathcal{O} \in S$ there is a witness output map $\text{Out}_X(\mathcal{O}; \cdot) : X \rightarrow \mathbb{R}$ such that

$$\text{Out}_{\mathcal{S}}(\mathcal{O}; \mathbf{p}) = \text{Out}_X(\mathcal{O}; r(\mathbf{p})) \quad (\forall \mathbf{p} \in P(\mathcal{S}), \forall \mathcal{O} \in S). \quad (1)$$

We call the presentation *sound* if $\mathbf{p} \sim_S \mathbf{p}' \Rightarrow r(\mathbf{p}) = r(\mathbf{p}')$, and *complete* if $r(\mathbf{p}) = r(\mathbf{p}') \Rightarrow \mathbf{p} \sim_S \mathbf{p}'$ (Definition 2.46).

Lemma 2.15 (Completeness of standing presentations). *Let (X, r, Out_X) be a standing presentation for scope S in the sense of Definition 2.14. Then the presentation is complete:*

$$r(\mathbf{p}) = r(\mathbf{p}') \implies \mathbf{p} \sim_S \mathbf{p}'.$$

Construction witness. Assume $r(\mathbf{p}) = r(\mathbf{p}')$. Then for every $\mathcal{O} \in S$, the factorization condition (1) gives

$$\text{Out}_{\mathcal{S}}(\mathcal{O}; \mathbf{p}) = \text{Out}_X(\mathcal{O}; r(\mathbf{p})) = \text{Out}_X(\mathcal{O}; r(\mathbf{p}')) = \text{Out}_{\mathcal{S}}(\mathcal{O}; \mathbf{p}').$$

Hence $\mathbf{p} \sim_S \mathbf{p}'$ by Definition 2.46. □

Lemma 2.16 (Soundness forced by stable identity fixation). *Assume minimal coherence for scope S (Definition 2.1). Let (X, r, Out_X) be a standing presentation for S in the sense of Definition 2.14. Then the presentation is sound:*

$$\mathbf{p} \sim_S \mathbf{p}' \implies r(\mathbf{p}) = r(\mathbf{p}').$$

Construction witness. Assume $\mathbf{p} \sim_S \mathbf{p}'$. Then for every $\mathcal{O} \in S$, $\text{Out}_{\mathcal{S}}(\mathcal{O}; \mathbf{p}) = \text{Out}_{\mathcal{S}}(\mathcal{O}; \mathbf{p}')$. By (1) this implies $\text{Out}_X(\mathcal{O}; r(\mathbf{p})) = \text{Out}_X(\mathcal{O}; r(\mathbf{p}'))$ for all $\mathcal{O} \in S$.

If $r(\mathbf{p}) \neq r(\mathbf{p}')$, then X distinguishes two putative “standing states” that are indistinguishable by *every* admissible S -admitted test. Maintaining such an unwitnessed distinction as identity-bearing violates stable identity fixation (MC1) under the fail-closed discipline: it either (i) privileges the particular r -labeling as identity-grounding (representational privilege), or (ii) imports extra unwitnessed structure to remember which label one is in (hidden load). Both are standing-failure modes (Definition 2.53). Therefore $r(\mathbf{p}) = r(\mathbf{p}')$ and the presentation is sound. □

Lemma 2.17 (Quotient universal property). *Let $q : P(\mathcal{S}) \rightarrow P(\mathcal{S})/\sim_S$ be the quotient map. For any set Y and any function $F : P(\mathcal{S}) \rightarrow Y$ that is constant on $\sim_S$ -classes (i.e. $\mathbf{p} \sim_S \mathbf{p}' \Rightarrow F(\mathbf{p}) = F(\mathbf{p}')$), there exists a unique function $\tilde{F} : P(\mathcal{S})/\sim_S \rightarrow Y$ such that*

$$F = \tilde{F} \circ q.$$

Construction witness. Define $\tilde{F}([\mathbf{p}]) := F(\mathbf{p})$. This is well-defined precisely because F is constant on $\sim_S$ -classes. Uniqueness follows because q is surjective: if $F = \hat{F} \circ q$ also, then for any $[\mathbf{p}]$, $\hat{F}([\mathbf{p}]) = \hat{F}(q(\mathbf{p})) = F(\mathbf{p}) = \tilde{F}(q(\mathbf{p})) = \tilde{F}([\mathbf{p}])$. $\square$

Theorem 2.18 (Admissibility constraint on uniqueness of the standing quotient). *Assume minimal coherence for scope S (Definition 2.1) inside the declared load and fixed-domain comparison class of Definition 1.8. Let (X, r, Out_X) be any standing presentation for S in the sense of Definition 2.14. Then r is automatically sound and complete and induces a unique bijection*

$$\phi : P(\mathcal{S})/\sim_S \longrightarrow X, \quad \phi([\mathbf{p}]) = r(\mathbf{p}).$$

In particular, any construction sequence that uses a non-isomorphic same-scope standing state space collapses: the standing quotient $P(\mathcal{S})/\sim_S$ is the unique standing state space for S (up to unique isomorphism) compatible with stable identity fixation and scope-indexed witness output invariance.

Construction witness. By Lemmas 2.15 and 2.16, the presentation is complete and sound. Define $\phi([\mathbf{p}]) := r(\mathbf{p})$.

Soundness implies ϕ is well-defined (it is constant on $\sim_S$ -classes). Completeness implies ϕ is injective: if $\phi([\mathbf{p}]) = \phi([\mathbf{p}'])$ then $r(\mathbf{p}) = r(\mathbf{p}')$ and Lemma 2.15 gives $\mathbf{p} \sim_S \mathbf{p}'$, hence $[\mathbf{p}] = [\mathbf{p}']$. Surjectivity holds because r is surjective.

Uniqueness of ϕ is exactly the quotient universal property (Lemma 2.17): r is constant on $\sim_S$ -classes (soundness), so it factors through q uniquely as $r = \phi \circ q$. $\square$

Realist coherence variant: continuation-separability as normal form. A recurring hostile move is to assert that **MC1** is “operationalist”: a realist may allow additional structure not distinguished by the base admitted test list. The meta-closure results above do not require that stronger metaphysical denial. What is required for *identity-bearing standing* is that alleged identity distinctions be *separable by some admissible continuation* in the declared scope. We therefore make the Myhill–Nerode analogy literal by defining admissible histories $\mathcal{H}_S$ as the closure of admitted tests under admissible interventions, and by exhibiting a canonical *reduction* of any standing presentation whose standing states are continuation-separated. In this sense **CR-Sep** is not an additional metaphysical axiom; it is the minimality (no redundant labels) normal form of any same-scope standing presentation.

Definition 2.19 (Admissible histories $\mathcal{H}_S$ as closure under composition). *Fix a scope declaration $S = (S_0, \mathcal{U}_S)$ and a skin $\mathcal{S}$ with parameter space $P(\mathcal{S})$, where S_0 is the admitted test family and $\mathcal{U}_S(\mathcal{S})$ is the monoid of admissible interventions/updates. Let $\mathcal{U}_S(\mathcal{S}) \subseteq \text{End}(P(\mathcal{S}))$ denote the monoid of admissible interventions/updates in scope S (it contains id and is closed under composition). For each admitted test $\mathcal{O} \in S_0$ define the associated test map $t_{\mathcal{O}} : P(\mathcal{S}) \rightarrow \mathbb{R}$ by*

$$t_{\mathcal{O}}(\mathbf{p}) := \text{Out}_{\mathcal{S}}(\mathcal{O}; \mathbf{p}).$$

Define the set of admissible histories/continuations to be the closure of the primitive tests under precomposition by admissible interventions:

$$\mathcal{H}_S(\mathcal{S}) := \{ t_{\mathcal{O}} \circ u : \mathcal{O} \in S_0, u \in \mathcal{U}_S(\mathcal{S}) \}.$$

Equivalently we may index a history by a pair $h = (u, \mathcal{O})$ with $u \in \mathcal{U}_S(\mathcal{S})$ and $\mathcal{O} \in S_0$ and write

$$\text{Out}_{\mathcal{S}}(h; \mathbf{p}) := \text{Out}_{\mathcal{S}}(\mathcal{O}; u(\mathbf{p})).$$

Define the continuation signature map

$$\Sigma_S : P(\mathcal{S}) \longrightarrow \mathbb{R}^{\mathcal{H}_S(\mathcal{S})}, \quad \Sigma_S(\mathbf{p}) := (\text{Out}_S(h; \mathbf{p}))_{h \in \mathcal{H}_S(\mathcal{S})}.$$

When $\mathcal{S}$ is fixed we write $\mathcal{H}_S := \mathcal{H}_S(\mathcal{S})$. We call a scope S history-saturated (relative to $\mathcal{U}_S$) if it already contains all composite tests $t_{\mathcal{O}} \circ u$ as admitted tests. In that case Definition 2.46 may be read as quantifying over $\mathcal{H}_S$, and the present notation makes the Myhill–Nerode “future-distinguishability” content explicit rather than rhetorical.

Lemma 2.20 (Kernel equivalence is the maximal update-stable refinement of primitive witness output equivalence). *Fix a scope declaration $S = (S_0, \mathcal{U}_S)$ and a skin $\mathcal{S}$. Define the primitive (snapshot) witness output-equivalence relation $\sim_0$ on $P(\mathcal{S})$ by*

$$\mathbf{p} \sim_0 \mathbf{p}' \iff \text{Out}_S(\mathcal{O}; \mathbf{p}) = \text{Out}_S(\mathcal{O}; \mathbf{p}') \quad \forall \mathcal{O} \in S_0.$$

Define the continuation (kernel) equivalence relation $\sim_{\text{ker}}$ by

$$\mathbf{p} \sim_{\text{ker}} \mathbf{p}' \iff \Sigma_S(\mathbf{p}) = \Sigma_S(\mathbf{p}'),$$

where Σ_S is the continuation-signature map of Definition 2.19. Equivalently, $\mathbf{p} \sim_{\text{ker}} \mathbf{p}'$ iff $\text{Out}_S(\mathcal{O}; u(\mathbf{p})) = \text{Out}_S(\mathcal{O}; u(\mathbf{p}'))$ for all $\mathcal{O} \in S_0$ and all $u \in \mathcal{U}_S(\mathcal{S})$.

Then:

- (i) $\sim_{\text{ker}}$ is a right-congruence for admissible update composition: if $\mathbf{p} \sim_{\text{ker}} \mathbf{p}'$ then $u(\mathbf{p}) \sim_{\text{ker}} u(\mathbf{p}')$ for all $u \in \mathcal{U}_S(\mathcal{S})$.
- (ii) $\sim_{\text{ker}}$ is the largest right-congruence contained in $\sim_0$. That is: if $\approx$ is any equivalence relation on $P(\mathcal{S})$ such that (a) $\mathbf{p} \approx \mathbf{p}' \Rightarrow \mathbf{p} \sim_0 \mathbf{p}'$ and (b) $\mathbf{p} \approx \mathbf{p}' \Rightarrow u(\mathbf{p}) \approx u(\mathbf{p}')$ for all $u \in \mathcal{U}_S(\mathcal{S})$, then $\mathbf{p} \approx \mathbf{p}' \Rightarrow \mathbf{p} \sim_{\text{ker}} \mathbf{p}'$.
- (iii) Consequently, if there exist $\mathbf{p}, \mathbf{p}'$ with $\mathbf{p} \sim_0 \mathbf{p}'$ but $\mathbf{p} \not\sim_{\text{ker}} \mathbf{p}'$ (kernel-inequivalent but snapshot-equivalent), then no quotient that identifies $\mathbf{p}$ and $\mathbf{p}'$ can admit a well-defined induced action of $\mathcal{U}_S(\mathcal{S})$ by $[\mathbf{p}] \mapsto [u(\mathbf{p})]$. Under (MC3) irreversible commitment (Definition 2.1), this ambiguity cannot be repaired post hoc by retroactive representative selection; any coherent identity notion compatible with composed updates must refine to $\sim_{\text{ker}}$. In a history-saturated scope, $\sim_{\text{ker}}$ coincides with $\sim_S$ (Definition 2.19), so identity collapses to the standing quotient.

Construction witness. (i) Assume $\mathbf{p} \sim_{\text{ker}} \mathbf{p}'$, i.e. $\Sigma_S(\mathbf{p}) = \Sigma_S(\mathbf{p}')$. Fix any $u_0 \in \mathcal{U}_S(\mathcal{S})$. For any history $h = (u, \mathcal{O}) \in \mathcal{H}_S$,

$$\text{Out}_S(h; u_0(\mathbf{p})) = \text{Out}_S(\mathcal{O}; u(u_0(\mathbf{p}))) = \text{Out}_S(\mathcal{O}; (u \circ u_0)(\mathbf{p})),$$

and similarly for $\mathbf{p}'$. Since $\mathcal{U}_S(\mathcal{S})$ is a monoid, $u \circ u_0 \in \mathcal{U}_S(\mathcal{S})$, so the equalities encoded in $\Sigma_S(\mathbf{p}) = \Sigma_S(\mathbf{p}')$ imply $\text{Out}_S(h; u_0(\mathbf{p})) = \text{Out}_S(h; u_0(\mathbf{p}'))$ for all $h \in \mathcal{H}_S$. Thus $\Sigma_S(u_0(\mathbf{p})) = \Sigma_S(u_0(\mathbf{p}'))$, i.e. $u_0(\mathbf{p}) \sim_{\text{ker}} u_0(\mathbf{p}')$.

(ii) Let $\approx$ satisfy (a)–(b) and assume $\mathbf{p} \approx \mathbf{p}'$. Then by (b), $u(\mathbf{p}) \approx u(\mathbf{p}')$ for all $u \in \mathcal{U}_S(\mathcal{S})$. By (a), this implies $u(\mathbf{p}) \sim_0 u(\mathbf{p}')$ for all u , i.e. for all $\mathcal{O} \in S_0$,

$$\text{Out}_S(\mathcal{O}; u(\mathbf{p})) = \text{Out}_S(\mathcal{O}; u(\mathbf{p}')).$$

By Definition 2.19 this is exactly $\Sigma_S(\mathbf{p}) = \Sigma_S(\mathbf{p}')$, hence $\mathbf{p} \sim_{\text{ker}} \mathbf{p}'$.

(iii) If $\mathbf{p} \sim_0 \mathbf{p}'$ but not $\mathbf{p} \sim_{\ker} \mathbf{p}'$, then by definition there exist $u \in \mathcal{U}_S(\mathcal{S})$ and $\mathcal{O} \in S_0$ with $\text{Out}_S(\mathcal{O}; u(\mathbf{p})) \neq \text{Out}_S(\mathcal{O}; u(\mathbf{p}'))$. Thus $u(\mathbf{p}) \not\sim_0 u(\mathbf{p}')$, so the induced map $[\mathbf{p}]_{\sim_0} \mapsto [u(\mathbf{p})]_{\sim_0}$ on $P(\mathcal{S})/\sim_0$ is not well-defined: the value depends on the chosen representative. Treating this dependence as standing-relevant is representational privilege, and attempting to “repair” it by retroactive representative selection violates (MC3) irreversible commitment. Therefore update-compatible identity must refine to $\sim_{\ker}$, which in a history-saturated scope agrees with $\sim_S$. $\square$

Definition 2.21 (Continuation-separability condition (CR-Sep)). *Let (X, r, Out_X) be a standing presentation for the history-saturated scope determined by Definition 2.19. Concretely: for every $h \in \mathcal{H}_S$ there is a witness output map $\text{Out}_X(h; \cdot) : X \rightarrow \mathbb{R}$ such that*

$$\text{Out}_S(h; \mathbf{p}) = \text{Out}_X(h; r(\mathbf{p})) \quad (\forall \mathbf{p} \in P(\mathcal{S})).$$

Define the continuation signature of a standing state $x \in X$ by

$$\Sigma_X(x) := (\text{Out}_X(h; x))_{h \in \mathcal{H}_S} \in \mathbb{R}^{\mathcal{H}_S}.$$

We say the presentation satisfies **CR-Sep** (continuation-separability) if Σ_X is injective: for any $x \neq x'$ there exists $h \in \mathcal{H}_S$ with $\text{Out}_X(h; x) \neq \text{Out}_X(h; x')$.

Definition 2.22 (Continuation-kernel reduction of a standing presentation). *Let (X, r, Out_X) be a standing presentation for the history-saturated scope determined by Definition 2.19. Define an equivalence relation $\approx_S$ on X by*

$$x \approx_S x' \iff \text{Out}_X(h; x) = \text{Out}_X(h; x') \text{ for all } h \in \mathcal{H}_S.$$

Let $\pi_X : X \rightarrow X^{\text{red}} := X/\approx_S$ be the quotient map. Define the reduced representation

$$r^{\text{red}} := \pi_X \circ r : P(\mathcal{S}) \rightarrow X^{\text{red}},$$

and define reduced witness outputs by

$$\text{Out}_{X^{\text{red}}}(h; [x]) := \text{Out}_X(h; x) \quad (h \in \mathcal{H}_S). \quad (2)$$

Lemma 2.23 (Reduction yields continuation-separability and forces soundness). *Let (X, r, Out_X) be any standing presentation for the history-saturated scope determined by Definition 2.19, and let $(X^{\text{red}}, r^{\text{red}}, \text{Out}_{X^{\text{red}}})$ be its reduction (Definition 2.22). Then:*

(i) $(X^{\text{red}}, r^{\text{red}}, \text{Out}_{X^{\text{red}}})$ is a standing presentation for S .

(ii) The reduced presentation is sound and complete.

(iii) The reduced presentation satisfies **CR-Sep** (Definition 2.21).

Consequently, **CR-Sep** can be taken without loss of generality: any same-scope standing presentation admits a canonical reduction whose standing states are continuation-separated.

Construction witness. (i) First note that $\approx_S$ is an equivalence relation (reflexive, symmetric, transitive) by the corresponding properties of equality of real numbers, quantified over $h \in \mathcal{H}_S$. To see that (2) is well-defined, suppose $[x] = [x']$ in X^{red} , i.e. $x \approx_S x'$. Then by definition $\text{Out}_X(h; x) = \text{Out}_X(h; x')$ for all h , so the right-hand side of (2) does not depend on the chosen

representative. Surjectivity of r^{red} follows because r and π_X are surjective. Finally, for any $\mathbf{p} \in P(\mathcal{S})$ and any $h \in \mathcal{H}_S$,

$$\text{Out}_S(h; \mathbf{p}) = \text{Out}_X(h; r(\mathbf{p})) = \text{Out}_{X^{\text{red}}}(h; \pi_X(r(\mathbf{p}))) = \text{Out}_{X^{\text{red}}}(h; r^{\text{red}}(\mathbf{p})),$$

so the factorization condition holds and $(X^{\text{red}}, r^{\text{red}}, \text{Out}_{X^{\text{red}}})$ is a standing presentation.

(ii) Soundness: assume $\mathbf{p} \sim_S \mathbf{p}'$. By Definition 2.46 (read as quantifying over histories via Definition 2.19), we have $\text{Out}_S(h; \mathbf{p}) = \text{Out}_S(h; \mathbf{p}')$ for all $h \in \mathcal{H}_S$. By factorization this implies $\text{Out}_X(h; r(\mathbf{p})) = \text{Out}_X(h; r(\mathbf{p}'))$ for all h , hence $r(\mathbf{p}) \approx_S r(\mathbf{p}')$ and therefore $r^{\text{red}}(\mathbf{p}) = \pi_X(r(\mathbf{p})) = \pi_X(r(\mathbf{p}')) = r^{\text{red}}(\mathbf{p}')$.

Completeness: assume $r^{\text{red}}(\mathbf{p}) = r^{\text{red}}(\mathbf{p}')$. Then $\pi_X(r(\mathbf{p})) = \pi_X(r(\mathbf{p}'))$, i.e. $r(\mathbf{p}) \approx_S r(\mathbf{p}')$, so $\text{Out}_X(h; r(\mathbf{p})) = \text{Out}_X(h; r(\mathbf{p}'))$ for all h and hence $\text{Out}_S(h; \mathbf{p}) = \text{Out}_S(h; \mathbf{p}')$ for all h by factorization. Thus $\mathbf{p} \sim_S \mathbf{p}'$.

(iii) Define the continuation signature on X^{red} by

$$\Sigma_{X^{\text{red}}}([x]) := (\text{Out}_{X^{\text{red}}}(h; [x]))_{h \in \mathcal{H}_S} = (\text{Out}_X(h; x))_{h \in \mathcal{H}_S}.$$

If $\Sigma_{X^{\text{red}}}([x]) = \Sigma_{X^{\text{red}}}([x'])$, then $\text{Out}_X(h; x) = \text{Out}_X(h; x')$ for all h , i.e. $x \approx_S x'$, hence $[x] = [x']$. So $\Sigma_{X^{\text{red}}}$ is injective and the reduced presentation satisfies CR-Sep. $\square$

Lemma 2.24 (CR-Sep implies soundness without invoking the weakest reading of MC1). *Let (X, r, Out_X) be a standing presentation for scope S and assume **CR-Sep** (Definition 2.21). Then the presentation is sound:*

$$\mathbf{p} \sim_S \mathbf{p}' \implies r(\mathbf{p}) = r(\mathbf{p}').$$

Construction witness. Assume $\mathbf{p} \sim_S \mathbf{p}'$. By Definition 2.14, for every $h \in \mathcal{H}_S$,

$$\text{Out}_S(h; \mathbf{p}) = \text{Out}_X(h; r(\mathbf{p})), \quad \text{Out}_S(h; \mathbf{p}') = \text{Out}_X(h; r(\mathbf{p}')).$$

Since $\mathbf{p} \sim_S \mathbf{p}'$ implies $\text{Out}_S(h; \mathbf{p}) = \text{Out}_S(h; \mathbf{p}')$ for all h (Definition 2.19), we have $\text{Out}_X(h; r(\mathbf{p})) = \text{Out}_X(h; r(\mathbf{p}'))$ for all $h \in \mathcal{H}_S$, i.e. $\Sigma_X(r(\mathbf{p})) = \Sigma_X(r(\mathbf{p}'))$. Injectivity of Σ_X (CR-Sep) gives $r(\mathbf{p}) = r(\mathbf{p}')$. $\square$

Theorem 2.25 (Admissibility constraint on realist identity: collapse to the standing quotient). *Fix a scope declaration $S = (S_0, \mathcal{U}_S)$ and a skin $\mathcal{S}$. Let (X, r, Out_X) be any standing presentation for the history-saturated scope determined by Definition 2.19, and let $(X^{\text{red}}, r^{\text{red}}, \text{Out}_{X^{\text{red}}})$ be its reduction (Definition 2.22). Then there exists a unique bijection*

$$\phi : P(\mathcal{S}) / \sim_S \longrightarrow X^{\text{red}}, \quad \phi([\mathbf{p}]) := r^{\text{red}}(\mathbf{p}).$$

In particular, the identity-bearing content of any same-scope standing presentation is (up to unique isomorphism) the standing quotient $P(\mathcal{S}) / \sim_S$; any additional same-scope structure in X beyond its reduction X^{red} is continuation-inert label redundancy.

Construction witness. By Lemma 2.23, the reduced representation r^{red} is sound and complete.

Well-definedness of ϕ : if $[\mathbf{p}] = [\mathbf{p}']$ in $P(\mathcal{S}) / \sim_S$, then $\mathbf{p} \sim_S \mathbf{p}'$ and soundness gives $r^{\text{red}}(\mathbf{p}) = r^{\text{red}}(\mathbf{p}')$.

Injectivity: if $\phi([\mathbf{p}]) = \phi([\mathbf{p}'])$, then $r^{\text{red}}(\mathbf{p}) = r^{\text{red}}(\mathbf{p}')$ and completeness implies $\mathbf{p} \sim_S \mathbf{p}'$, hence $[\mathbf{p}] = [\mathbf{p}']$.

Surjectivity: r^{red} is surjective because r and π_X are surjective.

Uniqueness: if $\psi : P(\mathcal{S}) / \sim_S \rightarrow X^{\text{red}}$ also satisfies $\psi([\mathbf{p}]) = r^{\text{red}}(\mathbf{p})$, then $\psi = \phi$ because the quotient map $q : P(\mathcal{S}) \rightarrow P(\mathcal{S}) / \sim_S$ is surjective (Lemma 2.17). $\square$

Remark 2.26 (Relation to Ad_S and NF_S). Theorem 2.25 shows that a realist can accept additional *unmodeled* structure only by either enlarging scope (admitting new histories/admitted tests) or adding new load (new witnesses); otherwise any same-scope identity refinement is eliminated by the reduction of Definition 2.22 (Lemma 2.23) and becomes haecceitistic. In particular, duality-related reorganizations that preserve all S -outputs cannot introduce new standing witnesses beyond the canonical normal form NF_S (Definition 6.10) and correspond to isomorphisms in the admissibility groupoid Ad_S (Theorem 2.13).

Corollary 2.27 (Transport completeness is forced by “same-scope” assertions). *Let $\mathcal{S}_A, \mathcal{S}_B$ be skins both used for scope S . If one asserts that $\mathcal{S}_A$ and $\mathcal{S}_B$ represent the same standing content for S (i.e. there exists a presentation X for which both factor as in (1)), then any admissible transport implementing that assertion must induce a bijection on standing quotients and is therefore transport-complete for S (Definition 2.36). Equivalently: within minimal coherence, agreement on all S -outputs plus “same-scope identity” already implies transport completeness; it is not an optional strengthening.*

Construction witness. By Theorem 2.18, each skin’s standing state space is uniquely $P(\mathcal{S}_\bullet)/\sim_S$ up to bijection. A transport implementing “same-scope identity” must therefore identify standing classes bijectively. If it failed transport completeness, some S -distinguishable pair in one skin would map to an S -indistinguishable pair in the other, contradicting the uniqueness of standing state spaces and violating stable identity fixation (MC1/MC3). Hence any same-scope transport must be transport-complete. $\square$

Theorem 2.28 (Admissibility constraint excluding standing-repair and canonical-selection operators). *Fix a scope S and assume minimal coherence. Any construction sequence using an asserted same-scope operator is constrained as follows: there is no admissible operator that, without enlarging the scope or adding new load, refines standing beyond $\sim_S$ or selects a standing-relevant canonical representative from each standing class. More precisely:*

- (i) *Any rule that distinguishes two points $\mathbf{p} \sim_S \mathbf{p}'$ as different identity-bearing states is a standing failure (Definition 2.53).*
- (ii) *Any “canonical representative” selector $s : P(\mathcal{S})/\sim_S \rightarrow P(\mathcal{S})$ (a section of the quotient map) may be used as a calculational convenience (gauge choice), but cannot be standing-relevant: treating $s([\mathbf{p}])$ as identity-grounding constitutes representational privilege and breaks (MC1).*
- (iii) *Consequently, typicality/measure/choice-selection operators that require a canonical representative or a non-invariant measure on representatives are inadmissible inside the same-scope interior; using them is an illicit scope transport (Definition 2.53(c)).*

Construction witness. (i) If $\mathbf{p} \sim_S \mathbf{p}'$ then by definition no S -witness distinguishes them. Declaring them different identity-bearing states therefore introduces hidden unwitnessed structure or privileged representation, i.e. a standing failure.

(ii) A section s chooses one representative per standing class. If s is treated as standing-relevant, then the identity of the constructed object depends on the choice of section, i.e. on a privileged representation. This violates (MC1). Therefore s can exist only as gauge convenience: any two choices of section are admissibly related and must not be used to generate new standing content.

(iii) A typicality or choice-selection operator requires additional structure beyond the standing class to define “most” or “selected” representatives (a measure, selection class, or selector). If that structure is not invariant under admissible redescription, it is representational privilege; if it is kept fixed anyway, it imports non-witnessed anchoring and is therefore an illicit scope transport. Either way it is inadmissible inside the same-scope interior. $\square$

Remark 2.29 (Common “standing repairs” and why they fail fast). Hostile reports often propose a meta-level “repair” to evade a closure result: add a selector, add a measure, add a completion operator, add a genericity/typicality rule, etc. Inside the present governance discipline these all reduce to the same typed failure: they attempt to introduce standing-relevant structure not fixed by $\sim_S$ and therefore not witnessed in scope S . Typical examples include:

- *Canonical representative selection* (choose a preferred element of each $[\mathbf{p}]$): violates Theorem 2.28(ii) unless treated as pure gauge convenience.
- *Typicality/measure postulates* on representatives (“most” points in a class): require a non-invariant measure or class-selection choice and are therefore representational privilege or illicit scope transport.
- *Choice-selection operators* (pick one completion/branch when multiple are S -indistinguishable): introduce hidden unwitnessed structure unless the choice is itself witnessed (scope change).
- *Completion/totalization operators* that assert they make partial data “total” without new load: they are either redundant relabelings of NF_S (no new standing) or alter witness content (scope change).
- *Genericity forcing / saturation / compactness-as-operator* repairs: these are meta-existence moves that select a privileged enlargement or completion; absent new standing witnesses they collapse back to representational privilege, and with witnesses they are scope changes.

The paper therefore fail-closes: such repairs are admissible only if they are exhibited as explicit load and scope changes with new witnesses, not as same-scope identity-preserving completions.

Theorem 2.30 (Admissibility constraint: same-lineage repair exhaustion). *Fix a scope S and a standing lineage inside the comparison class of Definition 1.8. Any construction sequence asserting same-lineage repair after a standing failure collapses. Every such repair attempt falls into one of five closed classes:*

- (i) temporal deferral: *admissibility is imposed only after a standing-relevant commitment;*
- (ii) silent redescription: *the standing target is changed without explicit admissible transport;*
- (iii) structural factorization: *admissibility of the whole is asserted from admissibility of parts;*
- (iv) localized containment: *standing failure is isolated to a region while the whole keeps standing;*
- (v) post-selection filtering: *inadmissible candidates are generated and standing is assigned only after selection or rejection.*

Conversely, all admissible same-lineage constructions must be contemporaneously admissible, explicitly identity-preserving under redescription, non-factorized at the standing boundary, globally standing-preserving across dependencies, and not repaired by post hoc selection.

Construction witness. The five classes exhaust the ways repair can be attempted without declaring a new scope: shift admissibility in time, shift the object silently, distribute standing across parts, localize the failure, or filter after production. Temporal deferral violates irreversible commitment; silent redescription violates identity preservation; structural factorization leaves interaction witnesses unadmitted or introduces a global witness after all; localized containment fails because standing-relevant dependencies are not erased by regional labels; and post-selection filtering cannot retroactively

confer standing on a construction path whose candidates were already generated. Any alleged sixth repair class either reduces to one of these five or introduces a new anchor/tensor item and is therefore a scope change by Theorem 2.51. $\square$

Corollary 2.31 (Duality reduction). *Let $\mathcal{S}_A, \mathcal{S}_B$ be any two skins (including duality-related reorganizations) that agree on all S -outputs and preserve standing under minimal coherence. Then they determine the same standing quotient up to isomorphism:*

$$P(\mathcal{S}_A)/\sim_S \cong P(\mathcal{S}_B)/\sim_S,$$

and hence cannot introduce additional invariant witnesses for S beyond those already present in the canonical normal form NF_S (Definition 6.10).

Construction witness. By Theorem 2.18, each skin's standing state space is uniquely its quotient. Agreement on all S -outputs implies the quotients have identical witness output rule, hence are isomorphic as standing presentations. Any purported new witness would refine standing beyond $\sim_S$, contradicting Theorem 2.28(i). $\square$

Theorem 2.32 (Admissibility constraint on maximality of Ad_S). *Fix a scope S inside the declared load and fixed-domain comparison class, and assume minimal coherence. Let Ad_S be the admissibility groupoid of same-scope identity-preserving transports (Theorem 2.13). Any construction sequence asserting a proper enlargement of Ad_S collapses unless it factors through the standing quotient. Equivalently, there is no proper enlargement of Ad_S that preserves the same scope S , introduces no new load, and remains minimally coherent. Any additional “operator” or “morphism” either:*

- (i) *factors through the standing quotient and therefore adds no new standing content (redundancy),*
or
- (ii) *changes the witness content and is therefore a scope change, or*
- (iii) *violates minimal coherence via a standing-failure mode.*

Construction witness. Let F be any additional same-scope map introduced on top of Ad_S . If F preserves S -outputs, then it cannot distinguish $\sim_S$ -equivalent points: if $\mathbf{p} \sim_S \mathbf{p}'$ but $F(\mathbf{p}) \neq F(\mathbf{p}')$, then F would introduce an identity-bearing distinction with no S -witness, contradicting Theorem 2.28(i). Thus F is constant on $\sim_S$ -classes. Let $q : P(\mathcal{S}) \rightarrow P(\mathcal{S})/\sim_S$ be the quotient map and define $\tilde{F}([\mathbf{p}]) := F(\mathbf{p})$. This is well-defined and satisfies $F = \tilde{F} \circ q$; uniqueness follows from surjectivity of q . Hence F adds no standing content beyond the quotient and is redundancy.

If F does not preserve S -outputs, then by definition it changes the witness content and is a scope change. If F purports to preserve S while distinguishing $\sim_S$ -equivalent points or selecting standing-relevant representatives, it violates Theorem 2.28 and hence minimal coherence. Therefore there is no proper same-scope standing-preserving extension of Ad_S . $\square$

2.3 Typed primitives

Definition 2.33 (Bookkeeping skin). *A bookkeeping skin $\mathcal{S}$ is a regime-scoped representational scheme: it specifies (i) a choice of variables/fields/parameters, and (ii) rules for computing witness outputs or invariant quantities in that regime.*

Definition 2.34 (Scope). *A scope S is a specification of:*

- (i) *a regime (energy/momentum range, on-shell/off-shell conditions, and degrees retained);*

- (ii) a class of admissible questions/admitted tests (e.g. low-energy CC contact amplitudes; pure QED scattering; on-shell high-energy longitudinal amplitudes);
- (iii) an admissible error discipline, if approximations are used (e.g. “up to $\mathcal{O}(p^2/m_W^2)$ ”).

All standing and witness assertions in this paper are scope-indexed: a parameter can be redundant for one scope and standing-relevant for a larger scope.

Definition 2.35 (Scope-equivalent skins). *Two skins $\mathcal{S}_A, \mathcal{S}_B$ are scope-equivalent for S if there exist admissible transports $T_{A \rightarrow B}$ and $T_{B \rightarrow A}$ such that, for every admitted test $\mathcal{O}$ admitted by S , the witness outputs agree:*

$$\text{Out}_A(\mathcal{O}; \mathbf{p}) = \text{Out}_B(\mathcal{O}; T_{A \rightarrow B}(\mathbf{p})) \quad \text{and} \quad \text{Out}_B(\mathcal{O}; \mathbf{q}) = \text{Out}_A(\mathcal{O}; T_{B \rightarrow A}(\mathbf{q})).$$

Definition 2.36 (Transport completeness). *An admissible transport $T : \mathcal{S}_A \rightarrow \mathcal{S}_B$ is transport-complete for scope S if $\mathcal{S}_B$ retains enough data to reproduce all standing-relevant distinctions of $\mathcal{S}_A$ that affect S -admitted tests. Equivalently: whenever two parameter points $\mathbf{p}, \mathbf{p}'$ in $\mathcal{S}_A$ yield different witness outputs for some $\mathcal{O} \in S$, their transports $T(\mathbf{p}), T(\mathbf{p}')$ differ in at least one invariant witness of $\mathcal{S}_B$.*

Remark 2.37 (EFT truncation is not admissible identity-preserving transport without an error envelope). An EFT may be an admissible skin for a scope S only if it is *matching-complete* for S : all operators whose coefficients affect S within the declared error tolerance are included, or else an explicit error envelope is carried as part of the witness bookkeeping. A truncated EFT presented as exact identity-preserving transport is inadmissible under the fail-closed discipline.

Remark 2.38 (No existence-by-assertion of admissible skins). Admissibility is fail-closed: the existence of an admissible skin, transport, or “further identity-preserving completion” is not a matter of plausibility. Any alleged additional admissible move must be exhibited by explicit construction and supplied with a transport certificate (Definition 2.57) inside the declared load and the declared scope. Bare existential assertions (“there may exist some other admissible skin not considered”) do not reopen the space.

Definition 2.39 (Standing). *Standing is the identity-fixation status of the quantities tracked by the bookkeeping. A datum has standing if it survives all admissible identity-preserving transformations and is transportable across admissible regime interfaces without importing extra load.*

Definition 2.40 (Admissible identity-preserving redescription). *An admissible identity-preserving redescription within a skin is any transformation that preserves (i) regime of applicability and (ii) standing of tracked objects. Examples in SMbookkeeping include: field redefinitions, unitary flavor-basis rotations, rephasings, gauge-fixing changes, and reparameterizations that leave invariant content unchanged.*

Definition 2.41 (Admissible transport). *A transport map $T : \mathcal{S}_A \rightarrow \mathcal{S}_B$ is an explicit rule mapping data in $\mathcal{S}_A$ to data in $\mathcal{S}_B$ that describe the same physical content in an overlapping regime (matching, integrating out, unbroken $\leftrightarrow$ broken transport, on-shell reduction, etc.). Transport is admissible if it preserves standing and introduces no new load outside Definition 1.2.*

Definition 2.42 (Invariant witness set). *Let $\mathcal{S}$ be a skin. An invariant witness set $\mathcal{W}(\mathcal{S})$ is a set of quantities (or equivalence classes) such that:*

- (i) each element of $\mathcal{W}(\mathcal{S})$ is invariant under every admissible redescription in $\mathcal{S}$;

(ii) for every admissible transport $T : \mathcal{S} \rightarrow \mathcal{S}'$, the transported standing-relevant content depends only on the images of $\mathcal{W}(\mathcal{S})$.

Definition 2.43 (Witness rank). *Fix a scope S . The witness rank for S is the minimal cardinality (finite or infinite) of independent real parameters required to specify a minimal invariant witness set sufficient to determine all S -outputs, with discrete skeleton labels held fixed. Independence is understood within the explicit certificate discipline of this paper: admissible witnesses and transports must be finitely presented and evaluable from the declared load, so pathological “oracle encodings” (e.g. hiding unbounded information in digits of a single real) are excluded as inadmissible. (Throughout, “rank” comparisons concern this continuous coordinate cardinality; discrete skeleton data are treated separately.)*

Remark 2.44 (Finite vs. functional witness rank). The witness rank may be infinite when the admitted scope requires genuinely functional data (e.g. a spectral density rather than finitely many numbers). In such cases, “rank” comparisons are understood as cardinal comparisons. See Proposition 3.7 for an explicit example in the electromagnetic sector.

Remark 2.45 (Finite-dimensional regular case). In the finite-parameter sectors treated in this paper, witness output maps and witness functions are analytic in the continuous parameters away from thresholds. In that regular setting the witness-rank bound of Theorem 2.54 matches ordinary parameter counting: if the standing quotient carries the structure of a finite-dimensional smooth manifold, any smooth injective witness map into $\mathbb{R}^m$ implies $\dim(P(\mathcal{S})/\sim_S) \leq m$ (invariance of domain / rank theorem). The cardinal formulation is retained only to accommodate scopes with functional witnesses (Remark 2.44).

Definition 2.46 (Standing equivalence for a scope). *Fix a scope S and a skin $\mathcal{S}$ with parameter space $P(\mathcal{S})$. For $\mathbf{p}, \mathbf{p}' \in P(\mathcal{S})$ write $\mathbf{p} \sim_S \mathbf{p}'$ iff, for every admitted test $\mathcal{O}$ admitted by S ,*

$$\text{Out}_S(\mathcal{O}; \mathbf{p}) = \text{Out}_S(\mathcal{O}; \mathbf{p}')$$

within the error discipline declared by S (Definition 2.34).

Definition 2.47 (Standing state space / quotient). *The standing state space of $\mathcal{S}$ for scope S is the quotient set $P(\mathcal{S})/\sim_S$. No canonical representative is assumed or required: only equivalence classes are used.*

Lemma 2.48 (Scope-equivalence induces an isomorphism of standing quotients). *Let $\mathcal{S}_A, \mathcal{S}_B$ be scope-equivalent skins for S (Definition 2.35). Assume there are admissible transports $T_{A \rightarrow B}$ and $T_{B \rightarrow A}$ that are transport-complete for S (Definition 2.36). Then $T_{A \rightarrow B}$ induces a bijection of standing state spaces*

$$P(\mathcal{S}_A)/\sim_S \cong P(\mathcal{S}_B)/\sim_S .$$

Construction witness. If $\mathbf{p} \sim_S \mathbf{p}'$ in $\mathcal{S}_A$, then for every $\mathcal{O} \in S$ we have $\text{Out}_A(\mathcal{O}; \mathbf{p}) = \text{Out}_A(\mathcal{O}; \mathbf{p}')$. By scope-equivalence, $\text{Out}_B(\mathcal{O}; T_{A \rightarrow B}(\mathbf{p})) = \text{Out}_A(\mathcal{O}; \mathbf{p})$ and similarly for $\mathbf{p}'$, hence $T_{A \rightarrow B}(\mathbf{p}) \sim_S T_{A \rightarrow B}(\mathbf{p}')$. Thus $T_{A \rightarrow B}$ is well-defined on the quotient.

If $\mathbf{p} \not\sim_S \mathbf{p}'$, then some $\mathcal{O} \in S$ differs in witness output. Transport completeness implies the transported points differ in standing-relevant witness content for S in $\mathcal{S}_B$, hence are not $\sim_S$ -equivalent. Therefore the induced map on quotients is injective. Applying the same construction sequence to $T_{B \rightarrow A}$ yields an inverse on quotients, hence bijectivity. $\square$

Definition 2.49 (Tensor load). *For a fixed scope, the tensor load is the minimal data required to fix all invariant witnesses across all admissible skins for that scope. Parameters not required to fix witnesses have no standing for that scope (redundancy) unless supported by additional invariant witnesses in every admissible skin.*

2.4 Anchor/tensor/skin role discipline

This subsection records the role typing imported into the present SM bookkeeping setting. It is not an additional dynamics or ontology. It is a fixed-domain sorting rule for components already present in the construction ledger.

Definition 2.50 (Role typing: anchor, tensor, skin). *Fix a scope S and the comparison class of Definition 1.8.*

- (i) *An anchor is a domain-fixing precondition: varying it changes definedness of the construction, the declared load, the admitted test family, the error envelope, or the licensed transport interface. In this paper the declared load, the chosen scope, the admissibility envelope, and the transport/error discipline are anchors.*
- (ii) *A tensor is standing-bearing quotient content: varying it, with anchors fixed, changes at least one invariant witness or admissible witness output. In this paper the tensor content is the witness ledger for the chosen scope, together with compatibility relations among witnesses.*
- (iii) *Skin is representational surplus: varying it, with anchors and tensor witnesses fixed, changes no standing quotient and no admissible witness output. Gauge choice, field basis, parameterization, notation, and matching presentation are skin when they are identity-preserving.*

Theorem 2.51 (Admissibility constraint on role exhaustion: no fourth role). *Within a fixed scope satisfying Definition 1.8, every component of a standing-bearing bookkeeping presentation is forced into exactly one of the roles in Definition 2.50. Any construction sequence asserting a fourth same-scope role collapses: the proposed component is either (i) anchor, because moving it changes the domain; (ii) tensor, because moving it changes a witness or admissible output; or (iii) skin, because moving it changes no standing content.*

Construction witness. Take any component C of a same-scope presentation. If varying C changes the declared load, scope, error envelope, definedness of transport, or admissibility interface, then C is anchor-typed. If those anchors are fixed and varying C changes some invariant witness or witness output, then C is tensor-typed. If neither definedness nor witness content changes, then C is skin-typed and carries no standing authority. These cases are exhaustive and mutually exclusive under the fail-closed discipline. A purported fourth role would have to be standing-relevant while changing neither domain nor witness content, which is hidden unwitnessed structure and hence a standing failure. $\square$

Corollary 2.52 (Selector, metadata, and scheduler collapse). *A canonical selector, representative-choice rule, metadata layer, ordering scheduler, or coherence functional cannot act as an additional same-scope authorizer. If it changes definedness or the admitted interface, it is anchor motion; if it changes witnesses or witness outputs, it is tensor load; if it changes neither, it is skin. In no case does it define a fourth standing-bearing role.*

Construction witness. Apply Theorem 2.51 to the proposed selector or scheduler. Authority without anchor or tensor effect is skin authority, which is excluded by quotient factorization; authority with such an effect is a declared scope/load change. Thus the only same-scope uses of these devices are calculational or notational. $\square$

2.5 Standing failures: the diagnostic taxonomy

Definition 2.53 (Standing-failure modes). *A purported independent degree of freedom is a standing failure if maintaining its independence requires at least one of:*

- (a) **hidden unwitnessed structure**: *dependence on data not represented by any invariant witness;*
- (b) **representational privilege**: *privileging a particular skin/basis/parameterization as identity-grounding;*
- (c) **illicit scope transport**: *importing structure justified only outside the declared scope;*
- (d) **compensating load/patching**: *adding fields/operators/sectors forbidden by the declared load;*
- (e) **silent boundary motion**: *changing the load, scope, witness family, transport/history structure, or role assignment while continuing to label the result “same-scope.”*

2.6 The engine theorem

Theorem 2.54 (Standing rank bound (witness counting)). *Fix a scope S (Definition 2.34), the declared load, and the fixed-domain comparison class of Definition 1.8. Let $\mathcal{S}_A, \mathcal{S}_B$ be scope-equivalent skins for S (Definition 2.35) and assume there exist admissible transports between them that are transport-complete for S (Definition 2.36). Let $P(\mathcal{S}_A)$ be parameterized by n real coordinates $\mathbf{p} = (p_1, \dots, p_n)$, with all discrete skeleton labels held fixed.*

Let m be the witness rank for S (Definition 2.43); equivalently, m is the minimal cardinality of real parameters needed to coordinatize the standing state space $P(\mathcal{S}_B)/\sim_S$ (Definition 2.47). Then $\mathcal{S}_A$ cannot carry more than m independent standing-relevant continuous degrees of freedom for S . In particular, if $n > m$ then $\mathbf{p}$ contains redundancy directions for S ; maintaining them as “standing” requires a standing failure (Definition 2.53).

Construction witness. By Lemma 2.48, scope-equivalence with transport completeness induces a bijection

$$P(\mathcal{S}_A)/\sim_S \cong P(\mathcal{S}_B)/\sim_S.$$

Hence it suffices to bound the standing coordinate cardinality in $\mathcal{S}_B$.

Let $\mathcal{W}_S(\mathcal{S}_B) = \{w_i\}_{i \in I}$ be a *minimal* invariant witness set determining all S -outputs, where I is an index set of cardinality $|I| = m$ (Definition 2.43). Define the *witness evaluation map*

$$w_B : P(\mathcal{S}_B) \longrightarrow \mathbb{R}^I, \quad w_B(\mathbf{p}) := (w_i(\mathbf{p}))_{i \in I},$$

where $\mathbb{R}^I$ denotes the set of functions $I \rightarrow \mathbb{R}$ (finite-dimensional vectors when I is finite).

Because $\mathcal{W}_S(\mathcal{S}_B)$ determines all S -outputs, for each $\mathcal{O} \in S$ there exists a function $F_{\mathcal{O}} : \mathbb{R}^I \rightarrow \mathbb{R}$ such that

$$\text{Out}_{\mathcal{S}_B}(\mathcal{O}; \mathbf{p}) = F_{\mathcal{O}}(w_B(\mathbf{p})) \quad (\forall \mathbf{p} \in P(\mathcal{S}_B)).$$

In particular, if $w_B(\mathbf{p}) = w_B(\mathbf{p}')$, then $\text{Out}_{\mathcal{S}_B}(\mathcal{O}; \mathbf{p}) = \text{Out}_{\mathcal{S}_B}(\mathcal{O}; \mathbf{p}')$ for all $\mathcal{O} \in S$, hence $\mathbf{p} \sim_S \mathbf{p}'$ by Definition 2.46. Thus w_B is constant on $\sim_S$ -classes and induces a well-defined map on the quotient,

$$\bar{w}_B : P(\mathcal{S}_B)/\sim_S \longrightarrow \mathbb{R}^I, \quad \bar{w}_B([\mathbf{p}]) := w_B(\mathbf{p}).$$

Moreover, $\bar{w}_B$ is injective: if $[\mathbf{p}] \neq [\mathbf{p}']$ then $\mathbf{p} \not\sim_S \mathbf{p}'$, so some $\mathcal{O} \in S$ has $\text{Out}_{\mathcal{S}_B}(\mathcal{O}; \mathbf{p}) \neq \text{Out}_{\mathcal{S}_B}(\mathcal{O}; \mathbf{p}')$. But then necessarily $w_B(\mathbf{p}) \neq w_B(\mathbf{p}')$, otherwise all S -outputs would coincide by the display above. Hence $\bar{w}_B([\mathbf{p}]) \neq \bar{w}_B([\mathbf{p}'])$.

Therefore $P(\mathcal{S}_B)/\sim_S$ embeds into $\mathbb{R}^I$ and can carry at most $|I| = m$ independent real coordinates (in the cardinal sense when m is infinite). The same bound holds for $\mathcal{S}_A$ because its standing quotient is bijective to that of $\mathcal{S}_B$.

If $\mathcal{S}_A$ is written with $n > m$ real coordinates, then the quotient map $P(\mathcal{S}_A) \rightarrow P(\mathcal{S}_A)/\sim_S$ cannot be injective: there exist distinct parameter points in $P(\mathcal{S}_A)$ representing the same standing class. Any coordinate direction along such fibers is redundancy for scope S . Maintaining such coordinates as “standing” would require either denying transport completeness/scope-equivalence, or importing additional standing structure (new witnesses, privileged parameterization, or hidden load), i.e. a standing-failure mode (Definition 2.53). $\square$

Remark 2.55 (Scope-indexed, not global). Theorem 2.54 does *not* assert that a parameter redundant for one scope is redundant in all scopes. Enlarging S (e.g. by including higher-energy admitted tests) can increase the minimal witness rank and restore standing to parameters previously redundant in a restricted regime.

Remark 2.56 (Non-novelty guardrail for witness counting). In finite-dimensional regular sectors, Theorem 2.54 recovers ordinary invariant-parameter counting. Its role here is not to relabel standard redundancy analysis, but to attach a fail-closed burden: extra same-scope standing requires both an invariant witness inside the fixed-domain comparison class and a transport-complete interface carrying that witness across skins. Without that witness/interface pair, the extra coordinate has skin status rather than tensor standing.

2.7 Transport certificates: how every collapse will be audited

Definition 2.57 (Transport certificate schema). A transport certificate for a collapse lemma consists of:

- (C1) explicit identification of the input skin $\mathcal{S}_A$ (variables and parameters);
- (C2) explicit identification of the output skin $\mathcal{S}_B$ (witness quantities);
- (C3) an explicit admissible transport $T : \mathcal{S}_A \rightarrow \mathcal{S}_B$ (matching, reduction, or redescription);
- (C4) an explicit computation of $\mathcal{W}(\mathcal{S}_B)$ and its rank;
- (C5) the rank comparison and the resulting redundancy/standing-failure diagnosis by Theorem 2.54.

3 Parameter normal form: fully worked construction witnesses and collapses

3.1 Charged-current collapse to G_F (fully worked)

Lemma 3.1 (Charged-current collapse in the contact-interaction scope). Let $S_{\text{CC}}^{(6)}$ be the charged-current contact scope defined by $p^2 \ll m_W^2$ and retention of the leading dimension-6 four-fermion operator, with corrections treated as $\mathcal{O}(p^2/m_W^2)$ (Definition 2.34). In $S_{\text{CC}}^{(6)}$, admissible transport from the renormalizable mediator skin to the four-fermion EFT skin exposes a single invariant witness G_F . Consequently, the pair (g_2, v) overcounts tensor load for this restricted scope: only the combination $G_F/\sqrt{2} = g_2^2/(8m_W^2)$ has standing as charged-current strength in $S_{\text{CC}}^{(6)}$.

Construction witness. We present a transport certificate in the sense of Definition 2.57.

(C1) Input skin $\mathcal{S}_W$ (mediator bookkeeping). In the electroweak gauge skin, charged currents couple to $W^\pm$ with coupling g_2 :

$$\mathcal{L}_{\text{int}}^{\text{CC}} = -\frac{g_2}{\sqrt{2}} \left(W_\mu^+ J_-^\mu + W_\mu^- J_+^\mu \right), \quad J_+^\mu := \bar{\psi}_1 \gamma^\mu P_L \psi_2, \quad (3)$$

where $P_L = (1 - \gamma_5)/2$ and $J_-^\mu = (J_+^\mu)^\dagger$.

(C2) Mediator mass from the Higgs kinetic term. Write the Higgs doublet in unitary gauge as

$$H(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}. \quad (4)$$

The gauge-covariant derivative is

$$D_\mu = \partial_\mu - i\frac{g_2}{2} \tau^a W_\mu^a - ig_1 Y B_\mu. \quad (5)$$

The Higgs kinetic term $(D_\mu H)^\dagger (D^\mu H)$ contains the charged gauge-boson mass term. Using $W_\mu^\pm = (W_\mu^1 \mp iW_\mu^2)/\sqrt{2}$ and the Pauli matrices, one finds the quadratic term

$$(D_\mu H)^\dagger (D^\mu H) \supset \frac{g_2^2 v^2}{4} W_\mu^+ W^{-\mu}. \quad (6)$$

Therefore

$$m_W^2 = \frac{g_2^2 v^2}{4}, \quad m_W = \frac{g_2 v}{2}. \quad (7)$$

(C3) Output skin $\mathcal{S}_F$ (low-energy four-fermion bookkeeping). In the low-energy charged-current EFT skin,

$$\mathcal{L}_{\text{EFT}}^{\text{CC}} = -\frac{G_F}{\sqrt{2}} (\bar{\psi}_1 \gamma^\mu (1 - \gamma_5) \psi_2) (\bar{\psi}_3 \gamma_\mu (1 - \gamma_5) \psi_4). \quad (8)$$

Here G_F is the *single* coupling witnessing charged-current strength in this regime.

(C4) Admissible transport $T : \mathcal{S}_W \rightarrow \mathcal{S}_F$ (integrate out W). In the regime $p^2 \ll m_W^2$, kinetic terms for W are subleading for matching and we may integrate out W at tree level by solving its algebraic equation of motion from the mass plus interaction terms:

$$\mathcal{L} \supset m_W^2 W_\mu^+ W^{-\mu} - \frac{g_2}{\sqrt{2}} (W_\mu^+ J_-^\mu + W_\mu^- J_+^\mu). \quad (9)$$

Varying with respect to W_μ^- gives

$$m_W^2 W^{+\mu} - \frac{g_2}{\sqrt{2}} J_+^\mu = 0 \quad \Rightarrow \quad W^{+\mu} = \frac{g_2}{\sqrt{2} m_W^2} J_+^\mu, \quad (10)$$

and similarly $W^{-\mu} = \frac{g_2}{\sqrt{2} m_W^2} J_-^\mu$. Substituting back yields the effective operator

$$\mathcal{L}_{\text{eff}}^{\text{CC}} = -\frac{g_2^2}{2m_W^2} J_+^\mu J_{-\mu}. \quad (11)$$

Now rewrite the left-handed currents in terms of $(1 - \gamma_5)$:

$$J_+^\mu J_{-\mu} = (\bar{\psi}_1 \gamma^\mu P_L \psi_2) (\bar{\psi}_3 \gamma_\mu P_L \psi_4) = \frac{1}{4} (\bar{\psi}_1 \gamma^\mu (1 - \gamma_5) \psi_2) (\bar{\psi}_3 \gamma_\mu (1 - \gamma_5) \psi_4). \quad (12)$$

Hence

$$\mathcal{L}_{\text{eff}}^{\text{CC}} = -\frac{g_2^2}{8m_W^2} (\bar{\psi}_1 \gamma^\mu (1 - \gamma_5) \psi_2) (\bar{\psi}_3 \gamma_\mu (1 - \gamma_5) \psi_4), \quad (13)$$

and matching to (8) gives

$$\frac{G_F}{\sqrt{2}} = \frac{g_2^2}{8m_W^2}. \quad (14)$$

Using (7) yields

$$\frac{G_F}{\sqrt{2}} = \frac{g_2^2}{8(g_2^2 v^2/4)} = \frac{1}{2v^2} \Rightarrow G_F = \frac{1}{\sqrt{2} v^2}. \quad (15)$$

(C5) Witness rank and collapse. The low-energy CC skin witnesses interaction strength by the single parameter G_F . The mediator skin uses (g_2, v) , but transport collapses all charged-current dependence to the single combination in (14) (equivalently (15)).

Therefore $\mathcal{W}(\mathcal{S}_F)$ has rank 1 while (g_2, v) has rank 2 in the mediator parameterization. By Theorem 2.54, (g_2, v) cannot represent two independent standing-relevant degrees for low-energy charged-current phenomena. The extra degree is redundancy for this scope. $\square$

Remark 3.2 (Scope warning for Lemma 3.1). Lemma 3.1 is deliberately restricted to the contact-interaction scope $S_{\text{CC}}^{(6)}$. If the scope is enlarged to resolve momentum dependence of the W propagator (or higher-dimension operators in the EFT), additional invariant witnesses appear (e.g. a witness for m_W), and the “collapse to G_F alone” no longer holds. This is an instance of Remark 2.55.

Proposition 3.3 (Worked example: enlarging $S_{\text{CC}}^{(6)}$ forces an additional witness). *Let $S_{\text{CC}}^{(8)}$ be the extended charged-current scope in which one resolves the leading momentum dependence of the W propagator: one retains the first correction of order p^2/m_W^2 beyond the contact limit, with residual errors treated as $\mathcal{O}(p^4/m_W^4)$. Equivalently, in the EFT skin one retains the dimension-6 four-fermion operator together with the leading derivative-dressed (dimension-8) correction required to reproduce the p^2/m_W^2 term. Then $S_{\text{CC}}^{(8)}$ contains two independent invariant witnesses, which may be taken as $\{G_F, m_W\}$ (or equivalently $\{C_6, C_8\}$, the coefficients of the dimension-6 and dimension-8 operators). In particular, m_W is an invariant witness in $S_{\text{CC}}^{(8)}$ and cannot be eliminated as redundancy in that enlarged scope.*

Construction witness. Work at tree level and consider a generic left-handed current–current process mediated by W exchange with momentum transfer p . The full mediator skin gives the amplitude factor (suppressing spinors and group factors already fixed in Lemma 3.1)

$$\mathcal{M}(p^2) \propto -\frac{g_2^2}{8} \frac{1}{m_W^2 - p^2} \mathcal{O}_6 + \dots, \quad (16)$$

where $\mathcal{O}_6$ denotes the Dirac/field structure

$$\mathcal{O}_6 := (\bar{\psi}_1 \gamma^\mu (1 - \gamma_5) \psi_2) (\bar{\psi}_3 \gamma_\mu (1 - \gamma_5) \psi_4),$$

and the ellipsis includes terms proportional to $p_\mu p_\nu / m_W^2$ in the massive-vector propagator. Those terms induce additional derivative operators but are suppressed by the same scale m_W ; they do not change the witness-counting point made here.

For $p^2 \ll m_W^2$, expand the propagator denominator:

$$\frac{1}{m_W^2 - p^2} = \frac{1}{m_W^2} \frac{1}{1 - p^2/m_W^2} = \frac{1}{m_W^2} \left(1 + \frac{p^2}{m_W^2} + \mathcal{O}\left(\frac{p^4}{m_W^4}\right) \right). \quad (17)$$

Substituting (17) into (16) yields

$$\mathcal{M}(p^2) \propto -\frac{g_2^2}{8m_W^2} \left(1 + \frac{p^2}{m_W^2} + \mathcal{O}\left(\frac{p^4}{m_W^4}\right) \right) \mathcal{O}_6. \quad (18)$$

Matching the p^0 term reproduces Lemma 3.1:

$$C_6 := \frac{G_F}{\sqrt{2}} = \frac{g_2^2}{8m_W^2}. \quad (19)$$

In the enlarged scope $S_{CC}^{(8)}$, one must also match the p^2 term. This requires an additional (local) operator in the EFT skin whose matrix element contributes a factor of p^2 relative to $\mathcal{O}_6$. One convenient representative (basis-dependent, but sufficient for witness extraction) is

$$\mathcal{O}_8 := (\bar{\psi}_1 \gamma^\mu (1 - \gamma_5) \psi_2) \square (\bar{\psi}_3 \gamma_\mu (1 - \gamma_5) \psi_4), \quad (20)$$

up to integration by parts and equivalent operator-basis choices. The corresponding Wilson coefficient is fixed by (18) as

$$C_8 = \frac{g_2^2}{8m_W^4} = \frac{C_6}{m_W^2} = \frac{G_F}{\sqrt{2}} \frac{1}{m_W^2}. \quad (21)$$

Hence the ratio of the two invariant coefficients is

$$\frac{C_8}{C_6} = \frac{1}{m_W^2}. \quad (22)$$

Within $S_{CC}^{(8)}$, this ratio is an *invariant witness*: it is extractable from the measured momentum dependence of charged-current amplitudes at order p^2/m_W^2 , and it is not determined by G_F alone. Therefore m_W (equivalently C_8) is standing-relevant in the enlarged scope and cannot be classified as redundancy there. This is precisely the mechanism described abstractly in Remark 2.55. $\square$

3.2 Electromagnetic collapse to e (fully worked)

Lemma 3.4 (Electromagnetic collapse in the pure-photon scope). *Let $S_{EM}^{(\gamma)}$ be the pure-photon electromagnetic scope in which Z exchange is integrated out (or kinematically irrelevant) and electromagnetic admitted tests are mediated by the massless gauge field A_μ alone (Definition 2.34). In $S_{EM}^{(\gamma)}$, admissible transport from electroweak gauge bookkeeping to the QED skin exposes a single invariant witness e . Consequently, treating (g_1, g_2) as independent electromagnetic couplings overcounts tensor load for $S_{EM}^{(\gamma)}$: only the combination $e = \frac{g_1 g_2}{\sqrt{g_1^2 + g_2^2}}$ has standing as electromagnetic strength in that scope.*

Construction witness. We again provide a transport certificate.

(C1) Input skin $\mathcal{S}_{EW}$ (neutral gauge bookkeeping). For a fermion field ψ with weak isospin generator T^3 and hypercharge Y (convention $Q = T^3 + Y$),

$$\mathcal{L} \supset \bar{\psi} \gamma^\mu (g_2 T^3 W_\mu^3 + g_1 Y B_\mu) \psi. \quad (23)$$

(C2) Neutral gauge-boson mass matrix from the Higgs kinetic term. Using the Higgs VEV as in Lemma 3.1, the neutral part of $(D_\mu H)^\dagger (D^\mu H)$ gives

$$(D_\mu H)^\dagger (D^\mu H) \supset \frac{v^2}{8} (g_2 W_\mu^3 - g_1 B_\mu)^2. \quad (24)$$

This quadratic form has one zero eigenvalue (massless photon) and one massive eigenstate (Z).

(C3) Diagonalization and definition of θ_W . Define the Weinberg angle by

$$\tan \theta_W = \frac{g_1}{g_2}, \quad \sin \theta_W = \frac{g_1}{\sqrt{g_1^2 + g_2^2}}, \quad \cos \theta_W = \frac{g_2}{\sqrt{g_1^2 + g_2^2}}. \quad (25)$$

Then the orthogonal rotation

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix} \quad (26)$$

diagonalizes (24), with A_μ massless.

(C4) Output skin S_{QED} (electromagnetic bookkeeping). The QED skin records electromagnetic interactions as

$$\mathcal{L}_{\text{QED}} \supset e A_\mu \bar{\psi} \gamma^\mu Q \psi, \quad (27)$$

with a single coupling e and charge operator Q .

(C5) Transport map and identification of the witness e . Insert (26) into (23). The coefficient of A_μ becomes

$$g_2 T^3 (\sin \theta_W) + g_1 Y (\cos \theta_W) = (g_2 \sin \theta_W) T^3 + (g_1 \cos \theta_W) Y. \quad (28)$$

Using (25), define

$$e := g_2 \sin \theta_W = g_1 \cos \theta_W = \frac{g_1 g_2}{\sqrt{g_1^2 + g_2^2}}. \quad (29)$$

Then the A_μ coupling is

$$e (T^3 + Y) = e Q, \quad (30)$$

which reproduces the QED form (27). Thus transport identifies *one* electromagnetic witness e .

(C6) Witness rank and collapse. The QED skin has a single coupling witness e for electromagnetic interaction strength. Electroweak bookkeeping uses (g_1, g_2) , but in the electromagnetic regime their only standing-relevant combination is (29). Therefore (g_1, g_2) overcounts tensor load for electromagnetic phenomena, and Theorem 2.54 forces collapse to e as the unique electromagnetic witness. $\square$

Remark 3.5 (Scope warning for Lemma 3.4). Lemma 3.4 is a statement about the pure-photon scope $S_{\text{EM}}^{(\gamma)}$. If the scope is enlarged to include Z exchange, parity-violating neutral currents, or energies where electroweak symmetry is effectively restored, then additional invariant witnesses (e.g. $\sin \theta_W$, m_Z) become standing-relevant and the collapse to e alone does not apply.

Proposition 3.6 (Worked example: enlarging $S_{\text{EM}}^{(\gamma)}$ to include Z exchange forces new witnesses).

Let $S_{\text{EM}}^{(\gamma)}$ be the pure-photon scope of Lemma 3.4. Define an enlarged scope $S_{\text{EM}}^{(\gamma Z, 6)}$ that additionally admits the leading ($q^2 \ll m_Z^2$) neutral-current correction from Z exchange in scattering of charged fermions, including polarization-dependent admitted tests. Then the witness set is strictly enlarged: beyond e , a new invariant witness is forced, and one may take it to be $s_W^2 \equiv \sin^2 \theta_W$. If the scope is further enlarged to $S_{\text{EM}}^{(\gamma Z, 8)}$ which resolves the first propagator correction at order q^2/m_Z^2 , then an additional invariant witness is forced, and one may take it to be m_Z .

Construction witness. We exhibit the mechanism explicitly, in the same certificate style as Proposition 3.3.

(E1) Z coupling and current structure. From the rotation (26), the Z coupling to a fermion with generators (T^3, Q) is

$$\mathcal{L} \supset g_Z Z_\mu \bar{\psi} \gamma^\mu (T^3 - s_W^2 Q) \psi, \quad g_Z := \frac{g^2}{\cos \theta_W} = \sqrt{g_1^2 + g_2^2}, \quad s_W \equiv \sin \theta_W. \quad (31)$$

In particular, for the electron ($Q_e = -1$) one has

$$g_L^e = T_{e_L}^3 - s_W^2 Q_e = -\frac{1}{2} + s_W^2, \quad g_R^e = T_{e_R}^3 - s_W^2 Q_e = 0 + s_W^2 = s_W^2. \quad (32)$$

(E2) Z -exchange amplitude and low-energy expansion. Consider t -channel exchange in (for definiteness) electron–electron scattering with momentum transfer q^2 (spacelike in elastic scattering). The Z -exchange contribution has the schematic form

$$\mathcal{M}_Z(q^2) \propto -\frac{g_Z^2}{m_Z^2 - q^2} [g_L^e (\bar{e}_L \gamma^\mu e_L) + g_R^e (\bar{e}_R \gamma^\mu e_R)]^2, \quad (33)$$

where m_Z is the Z mass. For $|q^2| \ll m_Z^2$, expand

$$\frac{1}{m_Z^2 - q^2} = \frac{1}{m_Z^2} \left(1 + \frac{q^2}{m_Z^2} + \mathcal{O}\left(\frac{q^4}{m_Z^4}\right) \right). \quad (34)$$

(E3) Matching at dimension six: s_W^2 becomes a witness. Keeping only the leading term of (34) (the scope $S_{\text{EM}}^{(\gamma Z, 6)}$), the Z exchange is represented in an EFT skin by local four-fermion operators. A convenient (chiral) basis is

$$\mathcal{O}_{6,ab}^{ee} := (\bar{e}_a \gamma^\mu e_a) (\bar{e}_b \gamma_\mu e_b), \quad a, b \in \{L, R\}, \quad (35)$$

with Wilson coefficients read off from (33) at order q^0/m_Z^2 :

$$C_{6,ab}^{ee} = \frac{g_Z^2}{m_Z^2} g_a^e g_b^e \quad (a, b \in \{L, R\}). \quad (36)$$

Crucially, the *relative* chiral strengths depend on s_W^2 through (32). For example,

$$\frac{C_{6,RR}^{ee}}{C_{6,LL}^{ee}} = \left(\frac{g_R^e}{g_L^e} \right)^2 = \left(\frac{s_W^2}{s_W^2 - \frac{1}{2}} \right)^2, \quad (37)$$

which is independent of the overall prefactor g_Z^2/m_Z^2 . Within $S_{\text{EM}}^{(\gamma Z, 6)}$ (where polarization-dependent admitted tests are admitted), the left–left and right–right neutral-current contact strengths are separately measurable in principle, so the ratio (37) is an *invariant witness*. It is *not* determined by the pure-photon witness e , since e does not fix s_W^2 (Lemma 3.4 only identifies the A_μ coupling). Thus enlarging $S_{\text{EM}}^{(\gamma)}$ to $S_{\text{EM}}^{(\gamma Z, 6)}$ forces at least one new witness; a convenient choice is s_W^2 .

(E4) Matching at dimension eight: m_Z becomes a witness. If the scope is enlarged further to $S_{\text{EM}}^{(\gamma Z, 8)}$ so that the q^2/m_Z^2 term in (34) is within the admitted accuracy, then additional (derivative) operators are required. A representative choice mirroring (20) is

$$\mathcal{O}_{8,ab}^{ee} := (\bar{e}_a \gamma^\mu e_a) \square (\bar{e}_b \gamma_\mu e_b) \quad (a, b \in \{L, R\}). \quad (38)$$

Matching the q^2 term yields

$$C_{8,ab}^{ee} = \frac{g_Z^2}{m_Z^4} g_a^e g_b^e. \quad (39)$$

Therefore, for any fixed chiral channel (a, b) ,

$$\frac{C_{8,ab}^{ee}}{C_{6,ab}^{ee}} = \frac{1}{m_Z^2}. \quad (40)$$

In $S_{\text{EM}}^{(\gamma Z, 8)}$, the momentum dependence of the neutral-current correction determines the ratio (40), hence m_Z is a standing-relevant invariant witness there.

Conclusion. The Z contribution is not “redundancy suppressed by heaviness”: it is simply outside $S_{\text{EM}}^{(\gamma)}$. Once the scope is enlarged to admit the corresponding admitted tests (contact neutral currents and/or their q^2 dependence), additional witnesses are forced by explicit matching, exactly as in Proposition 3.3. $\square$

Proposition 3.7 (Worked example: enlarging a pure-QED photon scope to include hadronic vacuum polarization forces new witnesses). *Let $S_{\text{EM}}^{(\gamma, \ell)}$ denote a pure-photon QED scope in which electromagnetic processes are mediated by the photon alone and vacuum-polarization effects are admitted only from the leptonic sector (so that the photon vacuum polarization is perturbatively computable within QED, given e and lepton masses). Define an enlarged scope $S_{\text{EM}}^{(\gamma, \ell + \text{had})}$ that additionally admits hadronic vacuum-polarization (HVP) corrections to photon exchange at the same perturbative order. Then the witness set is strictly enlarged: beyond e , additional invariant witnesses are forced. If the enlarged scope admits only a finite family of HVP-sensitive admitted tests, a finite witness set may be taken as the corresponding finite list of spectral integrals. If the enlarged scope admits energy-resolved hadronic production (so that $R_{\text{had}}(s)$ itself is an admitted test function of s) or otherwise a separating family of kernels, the witness is naturally represented by the hadronic spectral function $R_{\text{had}}(s)$ (equivalently the subtracted hadronic vacuum polarization $\Pi_{\text{had}}(q^2) - \Pi_{\text{had}}(0)$), which enters electromagnetic admitted tests through explicit dispersion integrals.*

Construction witness. We exhibit the scope enlargement explicitly and identify the new witnesses.

(H1) Photon vacuum polarization and the running coupling as a witness. Let j_{em}^μ be the electromagnetic current (defined so that the interaction is $\mathcal{L}_{\text{int}} = e A_\mu j_{\text{em}}^\mu$). Define the (transverse) current correlator

$$\Pi^{\mu\nu}(q) := i \int d^4x e^{iq \cdot x} \langle 0 | T j_{\text{em}}^\mu(x) j_{\text{em}}^\nu(0) | 0 \rangle = (q^\mu q^\nu - q^2 g^{\mu\nu}) \Pi(q^2). \quad (41)$$

The photon self-energy is $e^2 \Pi^{\mu\nu}$, so resumming vacuum-polarization insertions yields (in a convenient gauge)

$$D_{\mu\nu}(q) = \frac{-ig_{\mu\nu}}{q^2(1 - \widehat{\Pi}(q^2))} + \dots, \quad \widehat{\Pi}(q^2) := -e^2(\Pi(q^2) - \Pi(0)), \quad (42)$$

where the subtraction at $q^2 = 0$ fixes the on-shell normalization and “ $\dots$ ” are gauge terms that do not affect conserved currents. Accordingly, photon-mediated amplitudes acquire the replacement

$$\frac{e^2}{q^2} \mapsto \frac{e^2}{q^2(1 - \widehat{\Pi}(q^2))} \equiv \frac{e_{\text{eff}}^2(q^2)}{q^2}, \quad e_{\text{eff}}^2(q^2) := \frac{e^2}{1 - \widehat{\Pi}(q^2)}. \quad (43)$$

Thus, once vacuum polarization is admitted in the scope, the *function* $\widehat{\Pi}(q^2)$ (equivalently the running coupling $e_{\text{eff}}(q^2)$) is an invariant witness: it is extractable from momentum-dependent electromagnetic admitted tests and cannot be removed by admissible redescription.

(H2) Leptonic-only scope $S_{\text{EM}}^{(\gamma, \ell)}$. In the leptonic-only scope, $\widehat{\Pi}(q^2) = \widehat{\Pi}_\ell(q^2)$ is perturbatively computable in QED in terms of e and lepton masses. No additional, nonperturbative witness is required.

(H3) Enlarged scope $S_{\text{EM}}^{(\gamma, \ell + \text{had})}$ and extraction of a new witness. In the enlarged scope, the vacuum polarization splits as

$$\widehat{\Pi}(q^2) = \widehat{\Pi}_\ell(q^2) + \widehat{\Pi}_{\text{had}}(q^2). \quad (44)$$

Consider, for definiteness, $e^+e^- \rightarrow \mu^+\mu^-$ at center-of-mass energy $s = q^2$ in a regime where Z exchange is outside the scope (so the amplitude is photon-dominated). At leading order in the resummed vacuum polarization, the total cross section can be written as

$$\sigma_{\mu\mu}(s) = \sigma_{\mu\mu}^{(0)}(s) \left| \frac{1}{1 - \widehat{\Pi}(s)} \right|^2, \quad \sigma_{\mu\mu}^{(0)}(s) = \frac{4\pi\alpha^2}{3s}, \quad \alpha := \frac{e^2}{4\pi}. \quad (45)$$

Therefore the measured ratio

$$\left(\frac{\sigma_{\mu\mu}(s)}{\sigma_{\mu\mu}^{(0)}(s)} \right)^{1/2} - 1 \approx \text{Re } \widehat{\Pi}(s) \quad (46)$$

extracts $\widehat{\Pi}(s)$ as an invariant witness (to the admitted perturbative order). Since $\widehat{\Pi}_\ell(s)$ is computable in QED, the hadronic contribution

$$\widehat{\Pi}_{\text{had}}(s) = \widehat{\Pi}(s) - \widehat{\Pi}_\ell(s) \quad (47)$$

is itself an invariant witness in the enlarged scope. Crucially, $\widehat{\Pi}_{\text{had}}$ is *not* determined by e (or by the leptonic data) within the pure-QED skin.

(H4) Spectral representation: the hadronic witness is a dispersion integral. Analyticity of the correlator (41) implies a subtracted dispersion relation for the scalar function $\Pi(q^2)$:

$$\Pi(q^2) - \Pi(0) = \frac{q^2}{\pi} \int_{s_{\text{th}}}^{\infty} ds \frac{\text{Im } \Pi(s)}{s(s - q^2 - i\epsilon)}, \quad (48)$$

where s_{th} is the hadronic threshold. By the optical theorem, the hadronic spectral function is determined by the ratio

$$R_{\text{had}}(s) := \frac{\sigma(e^+e^- \rightarrow \text{hadrons}; s)}{\sigma_{\mu\mu}^{(0)}(s)}, \quad \sigma_{\mu\mu}^{(0)}(s) = \frac{4\pi\alpha^2}{3s}, \quad (49)$$

and one has the standard relation

$$R_{\text{had}}(s) = 12\pi \text{Im } \Pi_{\text{had}}(s). \quad (50)$$

Inserting (50) into (48) and multiplying by $-e^2$ yields the explicit spectral integral for the hadronic witness:

$$\widehat{\Pi}_{\text{had}}(q^2) = -e^2 \text{Re} (\Pi_{\text{had}}(q^2) - \Pi_{\text{had}}(0)) = -\frac{\alpha q^2}{3\pi} \text{PV} \int_{s_{\text{th}}}^{\infty} ds \frac{R_{\text{had}}(s)}{s(s - q^2)}. \quad (51)$$

Equation (51) exhibits the new witness explicitly as a *spectral integral*.

(H5) Kernel families and witness dimension. More generally, in any enlarged electromagnetic scope S admitting HVP-sensitive admitted tests whose hadronic contribution depends linearly on R_{had} , one may write

$$\Delta\mathcal{O}_K = \int_{s_{\text{th}}}^{\infty} ds K(s) R_{\text{had}}(s), \quad K \in \mathcal{K}_S, \quad (52)$$

for a set of admitted kernels $\mathcal{K}_S$ determined by which admitted tests are included in S . The kernel set $\mathcal{K}_S$ fixes the minimal witness requirement:

- (i) If $\mathcal{K}_S$ is finite, then a witness set may be taken as the finite list of numbers $\{\int ds K_i(s) R_{\text{had}}(s)\}$ associated with those kernels.
- (ii) If $\mathcal{K}_S$ *separates* spectral functions on the admissible class (i.e. whenever $R \neq R'$ there exists $K \in \mathcal{K}_S$ with $\int KR \neq \int KR'$), then $R_{\text{had}}(s)$ itself is forced as a standing-relevant witness (and the witness rank is infinite in the sense of Remark 2.44). In particular, if the scope admits the energy-resolved ratio $R_{\text{had}}(s)$ of (49) as an admitted test function of s (as is typical once HVP is admitted beyond a single integrated admitted test), then the separating condition holds trivially and a functional witness is unavoidable.

Thus, admitting HVP effects enlarges witness rank: at minimum it adds the required spectral integrals, and in the energy-resolved scope it forces the full function $R_{\text{had}}(s)$ (equivalently $\Pi_{\text{had}}(q^2)$), not merely the single coupling witness e .

Conclusion. Hadronic vacuum polarization is not “redundancy suppressed by nonperturbativity”; it is outside the leptonic-only scope. Once the scope is enlarged so that HVP-sensitive admitted tests are within the admitted accuracy, new invariant witnesses are forced and are naturally represented by spectral integrals such as (51). $\square$

Remark 3.8 (Scope interface: QED bookkeeping vs. QCD content). Proposition 3.7 is a scope statement about what a *pure-QED electromagnetic skin* must import once HVP effects are admitted. Within the full Standard Model declared load, the hadronic spectral function $R_{\text{had}}(s)$ is not posited as a new primitive; it is a witness of the QCD sector and can, in principle, be obtained either by direct measurement (via (49)) or by QCD computation. The point here is that it is *not determined by e alone* within the pure-photon QED skin, so enlarging the electromagnetic scope forces new witnesses and therefore cannot be misclassified as “redundancy.”

3.3 Yukawa–mass identity relation (fully worked, scope-explicit)

Lemma 3.9 (Yukawa–mass identity relation, per species). *Fix a fermion species f and work within the declared load. In the broken-phase skin, the Yukawa coupling y_f and the electroweak scale v do not represent independent standing-relevant degrees: they enter admitted tests only through*

$$m_f = \frac{y_f v}{\sqrt{2}}, \quad (53)$$

and (when Higgs interactions are kept) through the ratio m_f/v . Consequently, y_f has no independent standing beyond (m_f, v) , and in the low-energy mass-only scope the unique witness is m_f .

Construction witness. Write the Yukawa operator (schematically; H or $\tilde{H}$ as appropriate)

$$\mathcal{L}_Y = -y_f \bar{f}_L H f_R + \text{h.c.} \quad (54)$$

Insert $H = \frac{1}{\sqrt{2}}(0, v + h)^T$ to obtain

$$\mathcal{L}_Y = -\frac{y_f v}{\sqrt{2}} \bar{f} f - \frac{y_f}{\sqrt{2}} h \bar{f} f. \quad (55)$$

The first term is the fermion mass term, hence (53). The second term is the Higgs–fermion coupling; using (53) it becomes

$$-\frac{m_f}{v} h \bar{f} f. \quad (56)$$

Thus, in any broken-phase skin that retains the Higgs field h , the Yukawa data appear only as (m_f, v) rather than (y_f, v) . In any low-energy scope where Higgs exchange is integrated out (or irrelevant), standing-relevant fermion-sector content is witnessed by m_f alone (thresholds, propagators, kinematics), so y_f has no independent standing for that restricted scope.

Formally: the input parameterization (y_f, v) has rank 2, while the witness set for the mass-only scope has rank 1 (m_f). By Theorem 2.54, (y_f, v) cannot be two independent standing-relevant degrees in that scope. $\square$

Remark 3.10 (Scope warning). Lemma 3.9 does not deny that Higgs–fermion couplings are measurable. It states that, within the declared load, such measurements witness the *quotient combinations* m_f and m_f/v , not an additional standing degree beyond those witnesses for the chosen scope.

3.4 QCD CP–odd collapse to $\bar{\theta}$ (fully worked)

Lemma 3.11 (Strong CP: unique invariant $\bar{\theta}$). *In the QCD CP–odd sector, admissible chiral rephasing collapses apparent independence of the θ term and quark-mass phases. The unique standing-relevant invariant is*

$$\bar{\theta} := \theta + \arg \det M, \quad (57)$$

where M is the quark mass matrix.

Construction witness. We show explicitly how admissible rephasing acts, and compute the invariant.

(1) CP–odd operators in the ultraviolet skin. The relevant terms are

$$\mathcal{L} \supset -\bar{q}_R M q_L - \bar{q}_L M^\dagger q_R + \frac{g_s^2}{32\pi^2} \theta G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (58)$$

(2) Global axial rephasing of all quark flavors. Consider the (classically admissible) field redefinition

$$q \mapsto e^{i\alpha\gamma_5} q, \quad q_L \mapsto e^{-i\alpha} q_L, \quad q_R \mapsto e^{+i\alpha} q_R. \quad (59)$$

Under (59), the mass term transforms as

$$-\bar{q}_R M q_L \mapsto -\bar{q}_R e^{-2i\alpha} M q_L, \quad (60)$$

so

$$\arg \det M \mapsto \arg \det M - 2N_f \alpha, \quad (61)$$

where N_f is the number of quark flavors rotated.

(3) Anomalous Jacobian and the θ shift. In the quantum theory, the transformation (59) changes the path-integral measure (Fujikawa) and induces an additional term in the effective action:

$$\theta \mapsto \theta + 2N_f \alpha. \quad (62)$$

(Equivalently: the axial current has an anomaly proportional to $G\tilde{G}$, and the chiral rotation shifts the coefficient of $G\tilde{G}$.)

(4) Invariant combination. Combining (61) and (62) gives

$$\theta + \arg \det M \mapsto (\theta + 2N_f \alpha) + (\arg \det M - 2N_f \alpha) = \theta + \arg \det M. \quad (63)$$

Thus $\bar{\theta}$ in (57) is invariant under admissible chiral rephasing. Neither θ nor $\arg \det M$ is separately invariant; hence they do not have independent standing.

(5) Witness rank. Infrared CP-odd hadronic admitted tests depend on a single parameter in this sector; $\bar{\theta}$ is the unique candidate invariant under admissible rephasing. Therefore this scope has rank 1 witness content, forcing collapse to $\bar{\theta}$ as asserted. $\square$

3.5 Neutrino oscillation invariants (fully worked)

Lemma 3.12 (Oscillation witness set). *In the neutrino oscillation regime, transition probabilities depend only on:*

(i) *mass-squared differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$,*

(ii) *rephasing-invariant products $U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j}$.*

Absolute mass offsets and rephasing-removable phases are redundancy for oscillation phenomena.

Construction witness. Let U be the mixing matrix relating flavor and mass eigenstates:

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i}^* |\nu_i\rangle. \quad (64)$$

In the relativistic limit, propagation of $|\nu_i\rangle$ over baseline L with energy E yields a phase $\exp(-im_i^2 L/(2E))$ (up to an overall common phase that cancels in probabilities). Therefore the transition amplitude is

$$\mathcal{A}_{\alpha \rightarrow \beta} = \sum_i U_{\beta i} e^{-im_i^2 L/(2E)} U_{\alpha i}^*. \quad (65)$$

The probability is

$$\begin{aligned} P_{\alpha \rightarrow \beta} &= |\mathcal{A}_{\alpha \rightarrow \beta}|^2 = \sum_{i,j} U_{\beta i} U_{\alpha i}^* U_{\beta j}^* U_{\alpha j} \exp\left(-i \frac{(m_i^2 - m_j^2)L}{2E}\right) \\ &= \sum_{i,j} U_{\beta i} U_{\alpha i}^* U_{\beta j}^* U_{\alpha j} \exp\left(-i \frac{\Delta m_{ij}^2 L}{2E}\right). \end{aligned} \quad (66)$$

This exhibits the witness set directly: only Δm_{ij}^2 appear, and only the quartic products of U appear.

Now check admissible rephasing invariance. Rephase flavor and mass states by

$$|\nu_\alpha\rangle \mapsto e^{i\varphi_\alpha} |\nu_\alpha\rangle, \quad |\nu_i\rangle \mapsto e^{i\psi_i} |\nu_i\rangle. \quad (67)$$

Then $U_{\alpha i} \mapsto e^{i\varphi_\alpha} U_{\alpha i} e^{-i\psi_i}$, and the quartic product in (66) is invariant:

$$U_{\beta i} U_{\alpha i}^* U_{\beta j}^* U_{\alpha j} \mapsto (e^{i\varphi_\beta} U_{\beta i} e^{-i\psi_i})(e^{-i\varphi_\alpha} U_{\alpha i}^* e^{i\psi_i})(e^{-i\varphi_\beta} U_{\beta j}^* e^{i\psi_j})(e^{i\varphi_\alpha} U_{\alpha j} e^{-i\psi_j}) = U_{\beta i} U_{\alpha i}^* U_{\beta j}^* U_{\alpha j}. \quad (68)$$

Finally, shifting $m_i^2 \mapsto m_i^2 + c$ leaves all Δm_{ij}^2 unchanged and thus produces no new witness.

Therefore the oscillation regime witnesses only Δm_{ij}^2 and the rephasing-invariant quartic products, as asserted. $\square$

3.6 Longitudinal–Goldstone identity transport (worked construction-witness skeleton)

Lemma 3.13 (Longitudinal amplitude-class collapse). *In the electroweak high-energy regime $E \gg m_W$, on-shell transport identifies longitudinal vector boson amplitudes with Goldstone amplitudes up to $\mathcal{O}(m_W/E)$ corrections. Consequently, treating longitudinal polarizations and Goldstone modes as independent tensor degrees overcounts standing-relevant longitudinal content in that regime.*

Construction witness. We provide the minimal verifiable chain (the full BRST construction witness is standard; the steps below isolate the standing-relevant content).

(1) On-shell skin and longitudinal polarization. Consider an amplitude with an external massive gauge boson of momentum p^μ . A longitudinal polarization vector can be chosen to satisfy

$$\epsilon_L^\mu(p) = \frac{p^\mu}{m_W} + \mathcal{O}\left(\frac{m_W}{E}\right), \quad E \sim |p^0| \gg m_W. \quad (69)$$

(2) Ward/Slavnov–Taylor identity in a broken gauge theory. In R_ξ gauges, BRST invariance yields a Slavnov–Taylor identity relating the contraction of the (amputated) gauge amplitude with p_μ to the amplitude with the corresponding Goldstone field π . A construction sequence in the present conventions is given in Appendix C. For the standing-relevant content we need, it suffices to record the identity in the form

$$p_\mu \mathcal{M}^\mu(W^a(p), \dots) = i m_W \mathcal{M}(\pi^a(p), \dots), \quad (70)$$

up to terms that vanish for physical on-shell external states. The overall phase is a convention artifact of LSZ normalization and plays no role in the amplitude-class identification used here.

(3) High-energy reduction. Contract the gauge amplitude with the longitudinal polarization:

$$\mathcal{M}(W_L^a(p), \dots) = \epsilon_{L\mu}(p) \mathcal{M}^\mu(W^a(p), \dots) = \left(\frac{p_\mu}{m_W} + \mathcal{O}\left(\frac{m_W}{E}\right) \right) \mathcal{M}^\mu(W^a(p), \dots). \quad (71)$$

Using (70) gives

$$\mathcal{M}(W_L^a(p), \dots) = i \mathcal{M}(\pi^a(p), \dots) + \mathcal{O}\left(\frac{m_W}{E}\right). \quad (72)$$

(4) Standing consequence. In the high-energy on-shell regime, the amplitude class is witnessed by $\mathcal{M}$; the above shows that the longitudinal gauge-boson amplitude and Goldstone amplitude are the *same witness* up to a convention-dependent overall phase (here i) and controlled regime-suppressed corrections. Therefore counting W_L and π as independent tensor degrees overcounts standing-relevant content for that regime. $\square$

3.7 Parameter normal form (global statement)

Proposition 3.14 (Parameter normal form: witnesses exhaust standing). *Fix the declared load and a scope. A parameter has standing as an independent tensor degree for that scope only if it corresponds to an independent invariant witness across all admissible skins and survives admissible transport. Any parameter not so witnessed is redundancy for that scope, unless its retention can be justified without invoking standing-failure modes.*

4 Forced discrete skeleton: operators, anomalies, charges, FCNC

4.1 Operator existence is transport-invariant

Lemma 4.1 (Operator existence invariance). *Within the declared load, existence of a broken-phase mass pairing for a chiral fermion is equivalent to existence of a corresponding gauge-invariant Yukawa operator in the unbroken-phase skin. Absent the operator, no admissible redescription can create the pairing without forbidden patches.*

Construction witness. In the declared load, the renormalizable Lagrangian in the unbroken phase contains only gauge-invariant operators built from the available fields. Electroweak symmetry breaking is a *substitution* (choose a gauge and expand H around its VEV), followed by admissible field redefinitions (diagonalization, basis change). Such operations cannot create a term that is not the image of an existing gauge-invariant operator.

Therefore if a broken-phase mass term $\bar{f}f$ appears, it must arise from some gauge-invariant operator $\bar{f}_L H f_R$ or $\bar{f}_L \tilde{H} f_R$ already present (possibly after field relabeling). Conversely, if the corresponding gauge-invariant Yukawa operator is absent, producing the mass pairing requires adding (i) a new Higgs multiplet with different charges, (ii) new fermions enabling an effective pairing, or (iii) non-gauge-invariant/higher-dimensional patch operators. Each is forbidden by the declared load (Definition 1.2). Hence operator existence is transport-invariant. $\square$

4.2 Anomaly cancellation and Yukawa compatibility (full algebra)

We now show, explicitly, how the hypercharge structure is forced under the declared load. This subsection uses a *single consistent convention* throughout:

$$Q = T^3 + Y,$$

and the Higgs VEV is taken in the electrically neutral component of the doublet $H = (0, (v+h)/\sqrt{2})^T$, which fixes Y_H relative to T^3 by $Q(H^0) = 0$.

4.2.1 One-generation notation (right-handed hypercharges)

Work with one generation of fermions in the standard representation skeleton, written in terms of their usual chiralities:

Field	SU(3) _c × SU(2) _L rep	Hypercharge
Q_L (quark doublet)	(3, 2)	Y_Q
u_R (up singlet)	(3, 1)	Y_u
d_R (down singlet)	(3, 1)	Y_d
L_L (lepton doublet)	(1, 2)	Y_L
e_R (charged-lepton singlet)	(1, 1)	Y_e

The Higgs doublet has hypercharge Y_H and its conjugate $\tilde{H} = i\sigma^2 H^*$ has hypercharge $-Y_H$.

4.2.2 Yukawa gauge-invariance conditions

The renormalizable Yukawa terms in the declared load are

$$\mathcal{L}_Y \supset -\bar{Q}_L Y_u \tilde{H} u_R - \bar{Q}_L Y_d H d_R - \bar{L}_L Y_e H e_R + \text{h.c.} \quad (73)$$

Gauge invariance under $U(1)_Y$ fixes the hypercharge relations

$$Y_u = Y_Q + Y_H, \quad Y_d = Y_Q - Y_H, \quad Y_e = Y_L - Y_H. \quad (74)$$

(These are the familiar Standard-Model relations in the convention $Q = T^3 + Y$.)

4.2.3 Anomaly cancellation conditions (one generation)

Local anomaly constraints in chiral gauge theories are standard; see e.g. [4, 9]. Anomaly cancellation for a gauged $U(1)_Y$ requires vanishing of:

$$[\text{SU}(3)]^2 U(1) : \quad 2Y_Q - Y_u - Y_d = 0, \quad (75)$$

$$[\text{SU}(2)]^2 U(1) : \quad 3Y_Q + Y_L = 0, \quad (76)$$

$$[\text{grav}]^2 U(1) : \quad 6Y_Q + 2Y_L - 3Y_u - 3Y_d - Y_e = 0, \quad (77)$$

$$[U(1)]^3 : \quad 6Y_Q^3 + 2Y_L^3 - 3Y_u^3 - 3Y_d^3 - Y_e^3 = 0. \quad (78)$$

The minus signs for right-handed fields appear because anomalies are computed for left-handed Weyl fermions, so u_R, d_R, e_R contribute with opposite hypercharge.

Lemma 4.2 (Global $\text{SU}(2)$ (Witten) anomaly check). *In addition to the local triangle anomalies (75)–(78), a consistent $\text{SU}(2)_L$ gauge theory of left-handed Weyl fermions must have an even number of $\text{SU}(2)$ doublets (the Witten $\mathbb{Z}_2$ anomaly) [10]. In the minimal chiral skeleton, one generation contains 3 quark doublets (one per color) plus one lepton doublet, for a total of 4 doublets; hence the global $\text{SU}(2)$ anomaly cancels (and for n_g generations, $4n_g$ remains even).*

Construction witness. The nontrivial element of $\pi_4(\text{SU}(2)) \cong \mathbb{Z}_2$ can flip the sign of the fermion determinant for an odd number of left-handed doublets; non-collapse requires the sign be trivial. Counting $\text{SU}(2)$ doublets in the minimal skeleton gives $3 + 1 = 4$ per generation, hence even. $\square$

Proposition 4.3 (Constraint coupling inside the declared skeleton). *Within the minimal chiral skeleton:*

- (i) *If the Yukawa relations (74) hold, then the $[\text{SU}(3)]^2 U(1)$ anomaly (75) cancels identically.*
- (ii) *If the Yukawa relations (74) hold and the linear anomalies (76) and (77) vanish, then the cubic anomaly (78) vanishes automatically.*

Construction witness. (i) Substitute (74) into (75):

$$2Y_Q - (Y_Q + Y_H) - (Y_Q - Y_H) = 0.$$

(ii) Under (74), the mixed gravitational anomaly becomes

$$6Y_Q + 2Y_L - 3(Y_Q + Y_H) - 3(Y_Q - Y_H) - (Y_L - Y_H) = Y_L + Y_H.$$

Thus (77)=0 forces $Y_L = -Y_H$. Together with (76) (which gives $Y_L = -3Y_Q$) we obtain $Y_H = 3Y_Q$. Substituting $Y_L = -3Y_Q$ and $Y_H = 3Y_Q$ into (74) gives $Y_u = 4Y_Q$, $Y_d = -2Y_Q$, $Y_e = -6Y_Q$. A direct substitution of these relations into (78) gives 0 (see Appendix A for the explicit algebra). $\square$

Theorem 4.4 (Discrete-skeleton hypercharge collapse). *Within the declared load (minimal chiral skeleton + one Higgs doublet with the charged Yukawa operator content (73)), anomaly cancellation forces the one-generation hypercharges to the one-parameter family*

$$Y_L = -3Y_Q, \quad Y_e = -6Y_Q, \quad Y_H = 3Y_Q, \quad Y_u = 4Y_Q, \quad Y_d = -2Y_Q, \quad (79)$$

unique up to an overall normalization (absorbed into g_1) and an overall sign (charge conjugation).

Equivalently: the local anomaly equations alone determine $Y_L = -3Y_Q$, $Y_e = -6Y_Q$, and the unordered pair $\{Y_u, Y_d\} = \{4Y_Q, -2Y_Q\}$; the requirement that the charged lepton Yukawa term $-\bar{L}_L Y_e H e_R$ in (73) is gauge invariant fixes $Y_H = Y_L - Y_e = 3Y_Q$, and then the quark Yukawa relations (74) select the ordered assignment in (79) with u_R the singlet that couples via $\tilde{H}$.

Construction witness. Assume the anomaly conditions (75)–(78) with $Y_Q \neq 0$.

From (76) we obtain $Y_L = -3Y_Q$. From (75) we obtain $Y_u + Y_d = 2Y_Q$. From (77), using $Y_u + Y_d = 2Y_Q$, we obtain

$$0 = 6Y_Q + 2Y_L - 3(Y_u + Y_d) - Y_e = 6Y_Q + 2(-3Y_Q) - 3(2Y_Q) - Y_e = -6Y_Q - Y_e,$$

so $Y_e = -6Y_Q$.

Now write $Y_d = 2Y_Q - Y_u$ and substitute $Y_L = -3Y_Q$, $Y_e = -6Y_Q$ into the cubic constraint (78). Expanding (Appendix A) yields the quadratic equation

$$Y_u^2 - 2Y_Q Y_u - 8Y_Q^2 = 0, \quad (80)$$

and hence $Y_u \in \{4Y_Q, -2Y_Q\}$ with $Y_d = 2Y_Q - Y_u \in \{-2Y_Q, 4Y_Q\}$. Thus the anomaly equations fix the unordered pair $\{Y_u, Y_d\} = \{4Y_Q, -2Y_Q\}$.

Now impose existence of the charged lepton Yukawa operator $-\bar{L}_L Y_e H e_R$ in (73). Gauge invariance under $U(1)_Y$ requires the third relation in (74), $Y_e = Y_L - Y_H$, hence

$$Y_H = Y_L - Y_e = (-3Y_Q) - (-6Y_Q) = 3Y_Q. \quad (81)$$

With this Y_H , the quark Yukawa relations (74) force $Y_u = Y_Q + Y_H = 4Y_Q$ and $Y_d = Y_Q - Y_H = -2Y_Q$, yielding (79).

The overall normalization Y_Q is redundant (absorbed into g_1) and the overall sign corresponds to charge conjugation (Theorem 4.7). $\square$

Corollary 4.5 (Charge lattice rigidity). *Within the declared load, all anomaly-free hypercharge assignments are proportional to the standard solution; the only freedom is an overall normalization (absorbed into g_1) and a global sign. Consequently, once the redundancy of hypercharge normalization is fixed (Theorem 4.7), electric charge $Q = T^3 + Y$ is quantized on a discrete lattice.*

Remark 4.6 (Neutrino mass does not affect the forcing of hypercharge). Neutrino oscillation book-keeping (Section 3.5) can be accommodated either by adding gauge-singlet right-handed neutrinos (which are anomaly-inert if $Y_{\nu_R} = 0$) or by an effective Weinberg operator. Neither option alters the hypercharge forcing in Theorem 4.4 because the anomaly constraints and the charged Yukawa compatibility are already fixed by the charged chiral skeleton.

4.3 Hypercharge normalization and charge quantization

Theorem 4.7 (Hypercharge normalization collapse (redundancy elimination)). *Fix the declared load and the anomaly/Yukawa skeleton. The apparent continuous freedom to rescale hypercharge assignments is redundancy: it is removable by admissible redescription (a simultaneous rescaling of Y and g_1) and collapses to a canonical charge lattice choice, unique up to overall sign. Consequently, electric charge $Q = T^3 + Y$ is quantized on a discrete lattice fixed by the skeleton.*

Construction witness. Start in the unbroken gauge skin where hypercharge enters only through the product $g_1 Y$ in the covariant derivative. For any nonzero real λ , the simultaneous rescaling

$$Y \mapsto \lambda Y, \quad g_1 \mapsto g_1/\lambda \quad (82)$$

leaves $g_1 Y$ invariant and therefore leaves all gauge-phase interactions unchanged. Thus λ is *not* a standing-relevant degree; it is a redundancy in the gauge-phase parameterization.

In the broken-phase skin, the unbroken electromagnetic generator is

$$Q = T^3 + Y, \quad (83)$$

with T^3 fixed by the non-abelian representation content. Given Corollary 4.5, all hypercharges are proportional to a single normalization choice; choosing the representative in which the charge lattice is minimal (no common integer factor) fixes the redundancy (82). Only the global sign remains (charge conjugation).

Therefore hypercharge normalization is fixed up to sign as a redundancy elimination, and the resulting electric charges lie on a discrete lattice determined by the skeleton (hence are quantized). $\square$

4.4 Absence of tree-level FCNC (fully worked)

Theorem 4.8 (No tree-level flavor-changing neutral currents in the minimal load). *Within the declared load (single Higgs doublet; generation-universal gauge charges), neutral currents remain flavor-diagonal at tree level in the mass basis. Tree-level flavor-changing neutral currents (FCNC) require forbidden additional structure (extra Higgs multiplets, non-universal gauge generators, or compensating sectors).*

Construction witness. We show the diagonalization explicitly.

(1) Yukawa matrices and mass diagonalization. For n generations, the quark Yukawa terms are

$$\mathcal{L}_Y \supset -\bar{Q}_L Y_d H d_R - \bar{Q}_L Y_u \tilde{H} u_R + \text{h.c.}, \quad (84)$$

with complex $n \times n$ matrices Y_u, Y_d . After EWSB, $M_{u,d} = Y_{u,d} v/\sqrt{2}$. There exist unitary matrices $U_{uL}, U_{uR}, U_{dL}, U_{dR}$ such that

$$U_{uL}^\dagger M_u U_{uR} = \text{diag}(m_u, m_c, \dots), \quad U_{dL}^\dagger M_d U_{dR} = \text{diag}(m_d, m_s, \dots). \quad (85)$$

Define mass eigenstate fields

$$u'_L = U_{uL}^\dagger u_L, \quad u'_R = U_{uR}^\dagger u_R, \quad d'_L = U_{dL}^\dagger d_L, \quad d'_R = U_{dR}^\dagger d_R. \quad (86)$$

(2) Charged currents produce CKM. The left-handed charged current in the gauge basis is

$$\mathcal{L}_{CC} \supset -\frac{g_2}{\sqrt{2}} W_\mu^+ \bar{u}_L \gamma^\mu d_L + \text{h.c.} \quad (87)$$

In the mass basis,

$$\bar{u}_L \gamma^\mu d_L = \bar{u}'_L U_{uL} \gamma^\mu U_{dL}^\dagger d'_L = \bar{u}'_L \gamma^\mu (U_{uL}^\dagger U_{dL}) d'_L. \quad (88)$$

Thus the CKM matrix is $V \equiv U_{uL}^\dagger U_{dL}$ and charged currents are flavor-mixing.

(3) Neutral currents remain diagonal. The Z boson couples to neutral currents of the form

$$\mathcal{L}_{\text{NC}} \supset Z_\mu \bar{f} \gamma^\mu \left(g_L^f P_L + g_R^f P_R \right) f, \quad (89)$$

where $g_{L,R}^f$ depend only on the gauge charges of f (and are generation-universal in the declared load). For example, for left-handed up-type quarks $f = u$ one has

$$\bar{u}_L \gamma^\mu u_L = \bar{u}'_L U_{uL} \gamma^\mu U_{uL}^\dagger u'_L = \bar{u}'_L \gamma^\mu u'_L, \quad (90)$$

since U_{uL} is unitary. The same holds for d_L , u_R , d_R , and all leptons: a flavor-space unitary rotation cancels in $\bar{f} \gamma^\mu f$. Therefore all neutral currents in (89) are flavor-diagonal at tree level in the mass basis.

(4) Why FCNC require extra structure. Tree-level FCNC arise if (and essentially only if) the neutral current couplings are not proportional to the identity in flavor space *in the same basis that diagonalizes the mass matrices*. In the declared load, gauge charges are generation-universal and the neutral current generators are fixed, so the couplings are proportional to identity and remain diagonal after unitary rotations. To obtain non-universal neutral couplings one must add new Higgs multiplets with misaligned Yukawas, new gauge generators with non-universal charges, or compensating sectors—all forbidden by the declared load. $\square$

5 Mixing and rephasing: saturation and redundancy (fully shown)

5.1 Parameter counting and invariants

We state and prove the standard parameter-counting result, because it is the audit point for “mixing saturation” and “phase exhaustion”.

Theorem 5.1 (Physical mixing parameters as quotient invariants). *For n generations of quarks in the declared load, the Yukawa sector contains, after quotienting by admissible field redefinitions, the following physical invariants:*

- $2n$ quark masses;
- $\frac{n(n-1)}{2}$ mixing angles;
- $\frac{(n-1)(n-2)}{2}$ CP-odd phases (Dirac-type).

For $n = 3$, this is 6 masses, 3 angles, and 1 phase.

Construction witness. The two Yukawa matrices Y_u, Y_d are complex $n \times n$ matrices, hence contain $2n^2$ real parameters each, for a total of $4n^2$.

Admissible field redefinitions in the quark kinetic terms permit independent unitary rotations

$$Q_L \mapsto U_Q Q_L, \quad u_R \mapsto U_u u_R, \quad d_R \mapsto U_d d_R, \quad (91)$$

with $U_Q, U_u, U_d \in U(n)$, without changing gauge interactions (they are flavor-universal). These transformations act on Yukawas by

$$Y_u \mapsto U_Q Y_u U_u^\dagger, \quad Y_d \mapsto U_Q Y_d U_d^\dagger. \quad (92)$$

The group $U(n)^3$ has real dimension $3n^2$. However, one overall $U(1)$ corresponds to baryon number and does not change Y_u, Y_d (it acts trivially on both); therefore only $3n^2 - 1$ parameters can be removed. Hence the physical parameter count is

$$4n^2 - (3n^2 - 1) = n^2 + 1. \quad (93)$$

On the other hand, $n^2 + 1$ decomposes as

$$n^2 + 1 = 2n + \frac{n(n-1)}{2} + \frac{(n-1)(n-2)}{2}, \quad (94)$$

corresponding to $2n$ masses, $\frac{n(n-1)}{2}$ angles, and $\frac{(n-1)(n-2)}{2}$ phases. This matches the standard CKM parameterization count. For $n = 3$, one obtains 10 invariants: 6 masses, 3 angles, 1 phase. $\square$

5.2 Saturation and redundancy statements

Corollary 5.2 (Unitary mixing saturation). *In the declared load, all unitary mixing freedom is exhausted by admissible basis choice and Yukawa diagonalization. The CKM (and similarly PMNS, in the appropriate neutrino mass assumptions) is a quotient invariant of Yukawa operator content, not an additional independent tensor load.*

Remark 5.3 (Not “unphysical mixing”). Corollary 5.2 is often misread as a denial of physical mixing. It is not. It states that, within the declared load, mixing data are *quotient invariants* of the Yukawa operator content (misalignment of Y_u and Y_d) rather than an additional independent tensor load that can be specified on top of the Yukawas without changing witness content.

Corollary 5.4 (Rephasing redundancy exhaustion). *All rephasing freedom beyond the physical CP-odd quotient phases is redundancy. Any assertion of additional independent phases requires representational privilege or hidden structure and is therefore a standing failure (Definition 2.53).*

6 Exhaustion: no further admissible identity-preserving completion remains

6.1 What “identity-preserving completion” means here

An *identity-preserving completion* is any forced identification of an apparent parameter freedom as redundancy under admissible redescription and admissible transport, resulting in a smaller witness-determined tensor load for a fixed scope.

6.2 Admissibility envelope and a normal-form map

The word “exhaustion” is strong, so we explicitly delimit what is being exhausted.

Definition 6.1 (Admissibility envelope used here). *The admissibility envelope $\text{Ad}_{\mathcal{L}}$ used for the exhaustion assertion is the closure of the renormalizable gauge-phase skin under finite compositions of the following explicitly licensed moves inside the declared load:*

- (i) *admissible identity-preserving redescriptions (Definition 2.40): gauge-fixing changes, field redefinitions, flavor-basis rotations, and rephasings;*
- (ii) *broken $\leftrightarrow$ unbroken transport by explicit Higgs expansion and mass diagonalization;*

- (iii) regime transport by explicit matching/integration-out inside $\mathcal{L}$ (e.g. W -exchange $\rightarrow$ four-fermion, A/Z rotation $\rightarrow$ QED), with an explicit scope and error order when approximations are used (Definition 2.34 and Remark 2.37);
- (iv) on-shell reduction and high-energy limits with controlled corrections (e.g. the longitudinal-Goldstone transport).

We do not classify arbitrary skins outside $\text{Ad}_{\mathcal{L}}$; leaving the envelope is a scope change by Definition 1.5.

Theorem 6.2 (Certificate-maximality of $\text{Ad}_{\mathcal{L}}$ for fixed-domain standing-bearing transport). *Within the fixed-domain comparison class of Definition 1.8, $\text{Ad}_{\mathcal{L}}$ is maximal with respect to explicit finite same-domain construction moves that can carry standing without changing the declared load. More precisely, any proposed additional operator, transport, selector, presentation change, or completion M that preserves the declared load, fixed scope, error envelope, witness family, standing quotient, and continuation structure must fall into one of three cases:*

- (i) M is an identity-preserving redescription or a finite composition of the licensed load-internal transports listed in Definition 6.1;
- (ii) M factors uniquely through the normal form NF_S and therefore adds only skin-level bookkeeping;
- (iii) M moves an anchor, tensor witness, quotient, or continuation structure, and is therefore outside the same-domain comparison class rather than a counterexample to it.

Thus the envelope is not an arbitrary prefilter for the closure result. It is the certificate-maximal envelope for same-domain, standing-bearing, load-internal finite transport. Larger theories may extend it, but only by changing the domain whose closure is being audited. Conversely, any transformation outside $\text{Ad}_{\mathcal{L}}$ necessarily violates at least one of fixed-domain comparability, standing preservation, or transport completeness for the original scope, unless it is recertified as a finite same-domain composition or quotient factorization already covered by (i)–(ii).

Construction witness. Let M be an alleged same-domain extension of $\text{Ad}_{\mathcal{L}}$. If M changes the load, admitted tests, error discipline, transport/history composition, quotient relation, or witness ledger, it violates Definition 1.8 and is classified by Theorem 6.8. If it changes none of them, its action on standing classes is constant on $\sim_S$ -classes and factors through NF_S by Lemma 6.11. If it neither changes the domain nor factors through the quotient, then it distinguishes points without an invariant witness and collapses as hidden load or representational privilege. The only remaining case is a finite composition of the identity-preserving moves already licensed in Definition 6.1. This exhausts the same-domain enlargement routes available to an alleged counterexample. $\square$

Remark 6.3 (Outside-envelope transformations are not same-scope transformations). A transformation outside $\text{Ad}_{\mathcal{L}}$ may be mathematically legitimate and may define a valuable larger-domain construction. It is not, however, a same-scope transformation unless it preserves standing, transport completeness, and fixed-domain comparability. If it fails any one of these, it is outside the audit target; if it preserves all of them, Theorem 6.2 forces it either to factor through the normal form or to be expressible as a certificate-bearing finite same-domain composition. Thus outside- $\text{Ad}_{\mathcal{L}}$ status is not by itself an objection: it is a demand to specify which fixed-domain clause remains preserved and to provide the corresponding transport certificate.

Corollary 6.4 (No outside-envelope same-domain refuge). *A proposed transformation outside $\text{Ad}_{\mathcal{L}}$ cannot be kept as a same-domain standing-preserving operation by renaming it as a different formalism, dual presentation, nonlocal skin, latent variable, or histories-level carrier. It must either*

(i) supply a transport-complete certificate preserving Definition 1.8, in which case it factors through the normal form or belongs to the envelope by Theorem 6.2; or (ii) move load, scope, quotient, witness content, or continuation structure. Case (ii) is a larger-domain construction, not an internal counterexample.

Remark 6.5 (Exhaustion is typed, not global). Every exhaustion statement below is relative to $\text{Ad}_{\mathcal{L}}$, the declared scope family, and the fixed-domain comparison class. The result is intentionally not a classification of UV completions, nonrenormalizable new-load models, nonlocal reformulations outside the admitted transport envelope, or theories with different witness families. Such proposals may be valuable, but they are not same-scope counterexamples to an interior-closure assertion.

Remark 6.6 (Envelope maximality is same-domain maximality). The envelope $\text{Ad}_{\mathcal{L}}$ is not presented as a catalogue of all mathematical formulations. It is the maximal same-domain envelope generated by identity-preserving operations explicitly constructible from the declared load without changing the fixed-domain clauses of Definition 1.8. The formal statement is Theorem 6.2: adjoining an additional operation either adds no standing content because it factors through the standing normal form, or it changes an anchor, quotient, tensor witness, or continuation structure and therefore is not a same-scope enlargement of $\text{Ad}_{\mathcal{L}}$. Thus the asserted maximality is not “maximal by definition”; it is maximal under the construction certificate burden imposed by fixed-domain standing, transport completeness, and quotient preservation.

Remark 6.7 (Renormalization scheme and running are scope data). If a scope S includes variation in renormalization scale or scheme, then those choices are part of the scope specification (Definition 2.34) and any change of scale/scheme must be exhibited as an explicit transport (map of couplings and counterterms) inside the declared load. In this paper, couplings are treated as witnesses at a chosen renormalization point; renormalization-group running to another scale is a controlled transport, not an additional standing degree of freedom. Assertions that treat scheme dependence as an extra “physical parameter” without an explicit transport certificate are standing failures of type (b) or (c) (Definition 2.53).

Theorem 6.8 (No fifth same-scope locus for admissibility-bearing variation). *Within the fixed-domain comparison class (Definition 1.8) and the admissibility envelope $\text{Ad}_{\mathcal{L}}$, any same-scope standing-bearing variation must enter through one of four loci:*

- (i) *anchor motion: load, scope, error envelope, or transport-interface change;*
- (ii) *quotient motion: change of $\sim_S$ or of the standing quotient;*
- (iii) *tensor motion: change of invariant witness content or compatibility relations;*
- (iv) *continuation motion: change of the admissible history/composition structure used to compare skins.*

If none of (i)–(iv) changes, the variation factors through the canonical normal form and is skin. Therefore no fifth same-scope standing-bearing locus remains.

Construction witness. Let V be a proposed same-scope variation. If it changes load, scope, error discipline, or licensed transport, it moves an anchor and is outside the same fixed domain. If it changes which points are identified by $\sim_S$, it changes the standing quotient. If it leaves the quotient fixed but changes an invariant witness or a compatibility relation among witnesses, it is tensor motion. If it leaves witnesses fixed but changes which finite continuations are admissible, it changes the history/composition structure. If none of these occurs, every declared witness output and

continuation is unchanged on every standing class; by Lemma 6.11, V factors uniquely through NF_S and has no standing authority. This exhausts all same-scope positions available to a standing-bearing modification. $\square$

Definition 6.9 (Witness normal form for a scope). *Fix a scope S . A skin $\mathcal{S}$ is in witness normal form for S if its explicit parameterization consists only of (i) discrete-skeleton data required by S (e.g. hypercharge class, representation labels), and (ii) a minimal invariant witness set $\mathcal{W}_S(\mathcal{S})$ whose values determine all S -outputs.*

Definition 6.10 (Canonical standing normal form skin). *Fix a scope S and a skin $\mathcal{S}$ with parameter space $P(\mathcal{S})$. Define the canonical standing normal form $\text{NF}_S(\mathcal{S})$ to be the bookkeeping skin with parameter space*

$$P(\text{NF}_S(\mathcal{S})) := P(\mathcal{S}) / \sim_S$$

(Definition 2.47), and witness output rule inherited from $\mathcal{S}$ by

$$\text{Out}_{\text{NF}_S(\mathcal{S})}(\mathcal{O}; [\mathbf{p}]) := \text{Out}_{\mathcal{S}}(\mathcal{O}; \mathbf{p}) \quad (\mathcal{O} \in S). \quad (95)$$

This is well-defined because $\mathbf{p} \sim_S \mathbf{p}'$ implies equality of all S -outputs. No canonical representative is assumed: only standing classes $[\mathbf{p}]$ appear.

Lemma 6.11 (Compilation lemma: quotient transport and universal property). *Let S be a fixed scope and $\mathcal{S}$ any skin. Let $q : P(\mathcal{S}) \rightarrow P(\mathcal{S}) / \sim_S$ denote the quotient map. Then:*

- (i) *q defines an admissible transport $q : \mathcal{S} \rightarrow \text{NF}_S(\mathcal{S})$ that is transport-complete for S .*
- (ii) *(Universal property.) For any set X and any map $F : P(\mathcal{S}) \rightarrow X$ that is constant on $\sim_S$ -classes (i.e. $\mathbf{p} \sim_S \mathbf{p}' \Rightarrow F(\mathbf{p}) = F(\mathbf{p}')$), there exists a unique map $\tilde{F} : P(\mathcal{S}) / \sim_S \rightarrow X$ such that $F = \tilde{F} \circ q$.*

Construction witness. For (i), admissibility is immediate: q introduces no new load and preserves standing by construction. Transport completeness holds because if $\mathbf{p}, \mathbf{p}'$ yield different S -outputs, then $\mathbf{p} \not\sim_S \mathbf{p}'$, hence $q(\mathbf{p}) \neq q(\mathbf{p}')$; the normal form retains exactly the standing distinctions.

For (ii), define $\tilde{F}([\mathbf{p}]) := F(\mathbf{p})$. This is well-defined precisely because F is constant on $\sim_S$ -classes. Uniqueness follows because q is surjective. $\square$

Theorem 6.12 (Admissibility constraint on skin normal form completeness for a scope). *Fix a scope S .*

- (i) *(Terminal compression.) For any admissible transport $T : \mathcal{S} \rightarrow \mathcal{S}'$ that is scope-preserving for S , the induced map on parameter spaces factors uniquely through the quotient transport $q : \mathcal{S} \rightarrow \text{NF}_S(\mathcal{S})$. Equivalently: no admissible same-scope map can depend on more than the standing class $[\mathbf{p}]$.*
- (ii) *(Completeness.) Within the admissibility groupoid Ad_S (Theorem 2.13), the normal form is a complete invariant: if $\mathcal{S}_A \simeq \mathcal{S}_B$ in Ad_S , then $\text{NF}_S(\mathcal{S}_A) \cong \text{NF}_S(\mathcal{S}_B)$ as standing state spaces with witness output rule.*

Construction witness. (i) Scope-preserving admissible transport means: if $\mathbf{p} \sim_S \mathbf{p}'$ then the transported states are S -indistinguishable in $\mathcal{S}'$, hence $T(\mathbf{p})$ depends only on $[\mathbf{p}]$. Thus the underlying map on parameters is constant on $\sim_S$ -classes and factors uniquely through the quotient by Lemma 6.11(ii).

(ii) If $\mathcal{S}_A \simeq \mathcal{S}_B$ in Ad_S , then by definition there exist transport-complete admissible transports in both directions. Lemma 2.48 then gives a bijection of standing quotients preserving S -outputs, which is exactly an isomorphism $\text{NF}_S(\mathcal{S}_A) \cong \text{NF}_S(\mathcal{S}_B)$. $\square$

Lemma 6.13 (Existence/uniqueness of witness normal form inside $\text{Ad}_{\mathcal{L}}$). *For each scope treated in Sections 3–5, every skin in the envelope $\text{Ad}_{\mathcal{L}}$ can be brought, by a finite sequence of admissible redescrptions and admissible transports, to witness normal form for that scope. Any two witness normal forms for the same scope differ only by admissible redescription.*

Construction witness. Existence is constructive: Sections 3–5 explicitly exhibit the required transports, redundancy quotients, and matching maps for the relevant scope classes (contact CC, pure QED, CP-odd QCD, oscillations, high-energy longitudinal limits, and flavor-quotient mixing), thereby displaying explicit witness-coordinate presentations inside the envelope.

Uniqueness is a quotient statement: any two witness normal forms for the same S present the same standing state space and hence differ only by admissible redescription. Formally, the quotient $P(\mathcal{S})/\sim_S$ is canonical and any standing-preserving map factors through it uniquely (Lemma 6.11); there is no additional standing left to encode. $\square$

Theorem 6.14 (Admissibility constraint: identity-preserving exhaustion (relative to $\text{Ad}_{\mathcal{L}}$)). *Within the declared load, the fixed-domain comparison class of Definition 1.8, and the admissibility envelope $\text{Ad}_{\mathcal{L}}$ (Definition 6.1), for the scope family treated in Sections 3–5, any construction sequence asserting a further admissible identity-preserving completion beyond the following list collapses:*

- *the parameter normal-form collapses (Section 3),*
- *the forced discrete skeleton (Section 4),*
- *the mixing/rephasing quotient classification (Section 5).*

Any purported additional completion must either (i) leave $\text{Ad}_{\mathcal{L}}$ or the fixed-domain comparison class (a scope change), (ii) introduce new load, (iii) alter the witness family, standing quotient, or transport/history locus of Theorem 6.8, or (iv) invoke a standing-failure mode.

Construction witness. Fix one of the scopes treated in Sections 3–5 and let $\mathcal{S}$ be any skin in $\text{Ad}_{\mathcal{L}}$. By Lemma 6.13, $\mathcal{S}$ reduces (by explicit licensed moves) to a witness normal form presentation for S inside the envelope. Independently of which presentation is used, the standing content for S is the canonical quotient $P(\mathcal{S})/\sim_S$ (Definition 2.47), and $\text{NF}_S(\mathcal{S})$ is the canonical compression (Definition 6.10).

A further “identity-preserving completion” would have to identify (or remove) additional degrees of freedom while preserving all S -outputs. But any admissible scope-preserving transport out of $\mathcal{S}$ factors uniquely through the quotient transport to $\text{NF}_S(\mathcal{S})$ (Theorem 6.12(i)). Thus the quotient already performs the *maximal* standing-preserving identification: any stricter identification corresponds to a relation coarser than $\sim_S$, and therefore would identify two points that are S -distinguishable by definition of $\sim_S$. That changes an S -witness output and is therefore not a same-scope admissible completion.

Equivalently, in witness normal form every explicit parameter is either discrete skeleton data required by S or part of a *minimal* invariant witness set determining S -outputs. Removing a witness changes some S -witness output; keeping a parameter while denying its witness status contradicts Theorem 2.54 (or requires denying transport completeness), which is a standing failure.

Therefore no additional admissible completion exists *inside* $\text{Ad}_{\mathcal{L}}$ and Definition 1.8. The no-fifth-locus constraint (Theorem 6.8) and the envelope-maximality constraint (Theorem 6.2) exhaust the remaining same-domain escape route: a proposal that leaves the load/scope boundary, witness family, standing quotient, and transport/history structure fixed is skin; a proposal that changes one of them is no longer a same-scope completion. Any escape must import new transport moves or new structure (scope change/new load) or invoke standing-failure modes. $\square$

7 Capstone: the Standard-Model bookkeeping interior is closed (standalone statement)

Definition 7.1 (SM bookkeeping interior (in this paper)). *The SMbookkeeping interior is the identity-preserving class (under admissible redescription) of skins within the declared load whose discrete skeleton satisfies:*

- (i) *anomaly/Yukawa compatibility (Theorem 4.4);*
- (ii) *hypercharge/charge-lattice rigidity (Corollary 4.5 and Theorem 4.7);*
- (iii) *transport-closed parameter normal form (Proposition 3.14);*
- (iv) *flavor/phase quotient classification (Theorem 5.1).*

Theorem 7.2 (Interior closure and uniqueness (relative to the declared load and $\text{Ad}_{\mathcal{L}}$)). *Within the declared load, the fixed-domain comparison class of Definition 1.8, and the admissibility envelope $\text{Ad}_{\mathcal{L}}$, the SMbookkeeping interior is closed and unique up to admissible identity-preserving redescription. Any purported same-scope “extension” either (a) reduces to the same interior by admissible redescription, or (b) violates standing preservation by importing forbidden load, moving the scope, changing witness content, changing the standing quotient or continuation structure, privileging a representation, invoking a closed repair strategy, or performing illicit scope transport. This is not a uniqueness theorem over all quantum field theories; it is a relative closure theorem over the fixed-domain SMbookkeeping interior.*

Construction witness. The discrete skeleton is forced up to overall normalization/sign by Theorem 4.4 and Corollary 4.5, and normalization freedom is redundancy by Theorem 4.7. Within that skeleton, all apparent continuous freedoms not tied to invariant witnesses collapse under the transport certificates of Section 3, yielding the parameter normal form (Proposition 3.14). Mixing and phase freedoms are fully classified as quotient invariants (Theorem 5.1), and tree-level FCNC are excluded in the declared load (Theorem 4.8). Finally, Theorem 6.14, Theorem 6.8, Theorem 6.2, and Theorem 2.30 close the admissible identity-preserving completion space *inside* $\text{Ad}_{\mathcal{L}}$: no further admissible identity-preserving completion exists in the same fixed-domain comparison class.

Therefore any proposed modification that stays inside the declared load is either an admissible redescription of existing structure (hence reduces), or it introduces new operators/fields/sectors or privileges a representation to maintain standing—which is inadmissible by Definition 2.53 and Remark 1.4. Uniqueness is thus relative to the declared load and admissibility envelope. $\square$

Remark 7.3 (No global Standard-Model uniqueness assertion). Theorem 7.2 is a relative interior-closure theorem. It does not assert first-principles uniqueness among all possible fundamental theories, nor does it forbid larger-domain programs that add anchors, witnesses, boundary data, or transport structures. Such programs are typed as scope/load changes. If they preserve the fixed-domain comparison class instead, their standing content factors through the same normal form.

8 Certification: irreducible inputs and structural non-derivability

8.1 The full renormalizable scope and the witness ledger

The collapses in Sections 3–5 are *scope-indexed*: in restricted regimes (contact CC, pure QED, etc.) some couplings collapse to fewer witnesses. A hostile reader may then ask which data remain standing-relevant once the scope is enlarged back to the full renormalizable bookkeeping interior.

Definition 8.1 (Full renormalizable SMscope). *Let $S_{\text{SM}}^{\text{ren}}$ be the scope consisting of all witness outputs of the most general renormalizable Lagrangian built from the declared load (Definition 1.2), including processes sensitive to W , Z , Higgs exchange, QCD interactions, and flavor-changing charged currents. No low-energy truncation is assumed; basis/gauge choices are treated as admissible redescrptions.*

Proposition 8.2 (Witness normal form for $S_{\text{SM}}^{\text{ren}}$). *Fix the declared load and the full renormalizable scope $S_{\text{SM}}^{\text{ren}}$. Up to admissible redescription (gauge fixing, field redefinitions, flavor rotations, and rephasings), the standing-relevant data reduce to the following witness ledger:*

- (i) **Discrete skeleton:** gauge group and representation content (Definition 1.2), generation count n_g , and the hypercharge class fixed up to overall sign (Theorem 4.4 and Corollary 4.5).
- (ii) **Gauge-sector witnesses:** three gauge couplings (g_s, g_2, g_1) at a chosen renormalization scheme/scale (or any equivalent invariant reparameterization, e.g. $(\alpha_s, e, \sin^2 \theta_W)$ once the scope includes Z exchange).
- (iii) **Higgs-sector witnesses:** two independent invariants of the renormalizable Higgs potential, e.g. (v, m_h) (equivalently (μ^2, λ) in $V(H) = -\mu^2 H^\dagger H + \lambda(H^\dagger H)^2$).
- (iv) **Yukawa/flavor witnesses:** fermion mass eigenvalues (equivalently Yukawa singular values) together with the quotient mixing invariants (angles and CP phases, or equivalently Jarlskog-type invariants) as classified in Section 5.
- (v) **CP-odd strong-sector witness:** the unique invariant $\bar{\theta}$ (Lemma 3.11).
- (vi) **Neutrino-sector witnesses (if neutrino masses are included in scope):** mass-squared differences and rephasing-invariant mixing products as in Lemma 3.12, with the precise ledger depending on the Dirac/Majorana load choice (a scope/load choice, not fixed here).

No other continuous parameter in the renormalizable Lagrangian has independent standing for $S_{\text{SM}}^{\text{ren}}$.

Construction witness. Within the declared load, the most general renormalizable Lagrangian consists of:

- (a) gauge kinetic terms for $\text{SU}(3)_c$, $\text{SU}(2)_L$, and $\text{U}(1)_Y$ with couplings (g_s, g_2, g_1) ;
- (b) fermion kinetic terms with covariant derivatives fixed by the representation skeleton (no new parameters);
- (c) the Higgs kinetic term and a renormalizable potential $V(H) = -\mu^2 H^\dagger H + \lambda(H^\dagger H)^2$;
- (d) the Yukawa operator family (73) with complex matrices (Y_u, Y_d, Y_e) ;
- (e) the CP-odd topological density term with coefficient θ , whose standing-relevant invariant is $\bar{\theta}$ by Lemma 3.11.

There are no additional renormalizable gauge-invariant operators built from the declared fields.

Admissible redescrptions remove gauge-fixing parameters, field-redefinition artifacts, and the flavor-basis redundancies of Section 5. The hypercharge class is forced up to normalization/sign by Theorem 4.4 and Corollary 4.5, and the normalization freedom is redundancy by Theorem 4.7. The remaining independent invariant witnesses are exactly the items listed in (i)–(vi). $\square$

Remark 8.3 (Why this does not contradict the low-energy collapses). Proposition 8.2 concerns the full renormalizable scope $S_{\text{SM}}^{\text{ren}}$. In restricted scopes (e.g. $S_{\text{CC}}^{(6)}$ or $S_{\text{EM}}^{(\gamma)}$), transport collapses reduce witness rank, and fewer combinations (e.g. G_F or e) suffice. Enlarging the scope reintroduces additional witnesses (e.g. $m_W, m_Z, \sin^2 \theta_W$) exactly as in Remarks 3.2 and 3.5.

8.2 Relational identities vs. internal value fixation

Definition 8.4 (Relational identity vs. internal value fixation). *A relational identity is a standing-preserving relation among invariant witnesses that holds across all admissible skins in a scope (e.g. $m_W = g_2 v/2$, $G_F = 1/(\sqrt{2} v^2)$, or $\bar{\theta} = \theta + \arg \det M$). An internal value-fixation assertion is the stronger assertion that a specific numerical value of a witness parameter (e.g. $\alpha_s(m_Z)$, a Yukawa eigenvalue, or a mixing angle) is forced purely by interior constraints without importing any additional standing.*

Theorem 8.5 (Admissibility constraint: no same-interior fixation of numerical witness values). *Within the declared load, fixed-domain comparison class, and admissibility discipline of this paper, no internal value-fixation construction sequence of an irreducible witness parameter in Proposition 8.2 is admissible. This is not a statement against UV completions or boundary-condition programs; it is a same-interior exclusion. Any such same-interior assertion necessarily invokes at least one standing-failure mode (Definition 2.53)—representational privilege, illicit scope transport (e.g. measure or typicality postulates), compensating load, silent boundary motion, or a repair move already closed by Theorem 2.30.*

Construction witness. Let w be an irreducible invariant witness for $S_{\text{SM}}^{\text{ren}}$. By definition, w is invariant under admissible redescrptions, but its *value* is not fixed by the relations among witnesses exhibited inside the declared load: otherwise it would not be irreducible.

An internal value-fixation construction sequence would have to supply additional structure that selects $w = w_0$ without external boundary data. Any such selection must come from somewhere:

- If it is asserted as necessary by privileging a particular representation, parameterization, or basis as identity-grounding, then the assertion violates redescription stability and is a representational-privilege failure.
- If it is asserted as “generic”/“typical”/“measure-selected”, then the required measure/typicality operator is not generated by standing-preserving transport inside the declared load; importing it is illicit scope transport.
- If it is produced by adding new fields/operators/sectors or an external “selector” principle, that is compensating load forbidden by Definition 1.2.
- If it is asserted by delaying admissibility until a preferred value is selected, silently redefining the scope, composing local admissible fragments into a global value rule, localizing the failure to one sector, or filtering candidate values after construction, it falls under the repair-exhaustion closure of Theorem 2.30.

Role typing gives the same diagnosis: a numerical value selector is anchor if it changes the domain, tensor if it adds a new witness, and skin if it changes neither. These exhaust the ways to force a numerical value without empirical boundary input. Therefore any purported internal value-fixation construction sequence violates admissibility by invoking a standing-failure mode. $\square$

Remark 8.6 (Classification, not a ban). Theorem 8.5 is a classification statement: it does not forbid scope changes (e.g. UV completions) that introduce additional standing-relevant witnesses. It states only that the numerical values of the irreducible witnesses of the SMinterior are not derivable *inside* that interior without breaking the admissibility discipline.

Remark 8.7 (External value-selection programs are typed as larger-domain constructions). A UV completion, boundary-condition program, unification constraint, landscape measure, or high-scale

selector may attempt to fix numerical values of low-energy witnesses. The present theorem does not reject such programs; it types them. If the program adds a new witness, boundary datum, selector, measure, or transport rule, it has moved the anchor/tensor structure and therefore leaves the same-domain interior. If it adds none of these, it cannot select a numerical value without privileging skin, violating Theorem 6.2, or invoking a repair move closed by Theorem 2.30. The theorem therefore excludes only same-interior value fixation, not larger-domain value-selection programs.

9 Hostile-referee attack map (standalone)

The table below lists common attack moves and the exact result that must be denied to reopen the space. Any objection not listed must identify which hypothesis of the cited result is being rejected.

Objection / escape attempt	Must deny or else blocked by
“You are conflating redundancy with regime suppression / integrating-out. The witness theorem is not justified.”	Definitions 2.34–2.47, Lemma 2.48, Remark 2.37, and Theorem 2.54.
“Declared load is a hidden anchor: you assumed the SM field content.”	Remark 1.3, Section 1.6, and Proposition 1.6.
“You did not derive the gauge group / representation skeleton, so this is not a construction of the SM.”	Remark 1.3 (typed scope: closure relative to a fixed load).
“There may exist other admissible skins or further identity-preserving completions you did not consider.”	Remark 2.38, Definition 6.1, and Theorem 6.2 (construction requirement plus maximal same-domain envelope).
“ G_F matching is just a convention; g_2 and v are independent in CC physics.”	Lemma 3.1 (explicit matching (14)) <i>for scope</i> $S_{CC}^{(6)}$ and Remark 3.2 (scope warning).
“ e is not the only electromagnetic witness; g_1 and g_2 are separately physical in EM.”	Lemma 3.4 (rotation (26) and identification (29)) <i>for scope</i> $S_{EM}^{(\gamma)}$ and Remark 3.5.
“ θ and $\arg \det M$ are independent physical parameters.”	Lemma 3.11 (explicit shifts (61)–(62)).
“You overstate anomaly $\rightarrow$ Yukawa forcing / the algebra is inconsistent.”	Section 4.2, Proposition 4.3, and Theorem 4.4 (fully worked, convention-consistent forcing).
“Hypercharge normalization is a genuine extra continuous freedom.”	Theorem 4.7 (rescaling redundancy (82)).
“Mixing saturation denies physical mixing.”	Theorem 5.1 and Remark 5.3.
“Tree-level FCNC can appear without extra structure; your theorem is not new.”	Theorem 4.8 (explicit unitary cancellation; novelty here is the scope/standing role typing, not the cancellation itself).
“You assert determination of the numerical values of the remaining SM parameters.”	Definition 8.4, Theorem 8.5, and Remark 8.6.
“There are further identity-preserving completions left; your exhaustion assertion is hand-wavy.”	Remark 2.38, Definition 6.1, Lemma 6.13, and Theorem 6.14.
“This is tautological: the Standard Model load was assumed and the Standard Model was recovered.”	Section 1.3, Remark 1.3, and Remark 6.5; the target is relative interior closure, not external construction of the load.
“Admissibility is just empirical equivalence with new terminology.”	Definition 2.5, Remark 2.8, Proposition 2.6, Lemma 2.7, Lemma 2.20, and Theorem 2.30.

Objection / escape attempt	Must deny or else blocked by
“The exhaustion theorem only covers a pre-filtered class.”	Definition 6.1, Remark 6.5, Remark 6.6, and Theorem 6.2; outside-envelope moves are either skin-redundant after quotient factorization or scope/load/transport changes rather than counterexamples.
“ $\text{Ad}_{\mathcal{L}}$ is doing hidden selection work.”	Definition 1.8, Remark 6.6, Theorem 6.2, and Theorem 6.8; any additional same-domain operation either factors through normal form or moves an explicit locus.
“ $\text{Ad}_{\mathcal{L}}$ is not maximal; there may be further same-scope transformations not generated by your list.”	Theorem 6.2, Corollary 6.4, Lemma 6.11, and Remark 2.38; an extra transformation must either factor through normal form, move a fixed-domain locus, or supply a transport certificate.
“ $\text{Ad}_{\mathcal{L}}$ maximality is merely definitional, not necessary.”	Remark 1.10, Theorem 6.2, Corollary 6.4, and Theorem 6.8; outside-envelope standing requires preserved comparability, standing, and transport completeness, otherwise the domain has moved.
“Minimal coherence bakes in the conclusion.”	Definition 2.1, Remark 2.9, Theorem 2.13, and the Referee Construction-Witness Obligations in Section 2.2; a refutation must deny a named MC condition or exhibit a non-isomorphic standing presentation.
“Irreversibility is imposed; a histories or path-integral formalism need not accept it.”	Definition 2.1, Remark 2.4, Lemma 2.7, and Theorem 2.30; if committed records may be reindexed after use, stable same-domain identity has been abandoned.
“Standing is merely epistemic bookkeeping, not structural physics.”	Remark 2.11, Remark 2.12, Proposition 1.6, and Definition 1.8; broader-domain structure must enter as new load, new witnesses, or new scope.
“Latent structure may exist even if invisible in your scope.”	Remark 2.12, Proposition 1.6, and Theorem 1.11; invisible structure is skin unless it becomes a witness by scope/load/transport change.
“Latent structure becomes witnessable under an arbitrarily small scope extension.”	Remark 2.12 and Remark 1.12; any change in admitted tests or transport interface is a typed scope change, not a same-scope completion.
“External programs can fix numerical values, so the non-fixability theorem is false.”	Theorem 8.5, Remark 8.6, and Remark 8.7; such programs are typed as new load, boundary data, anchor motion, or same-domain selectors already blocked.

Objection / escape attempt	Must deny or else blocked by
“This ignores standard redundancy, EFT, algebraic-QFT, or duality literature.”	Section 1.4; the local calculations are standard, while the asserted contribution is the fixed-domain closure ledger.
“The witness-rank theorem is just ordinary parameter counting.”	Remark 2.56; ordinary counting is recovered in regular sectors, but extra same-scope standing requires a fixed-domain invariant witness plus a transport-complete interface.
“The title still implies global uniqueness of the Standard Model.”	Section 1.3, Remark 1.12, and Remark 7.3; the project is explicitly a fixed-load relative-closure ledger.

Additional fixed-domain hardening targets. The following escape attempts are blocked by the fixed-domain role and repair closures above:

- “The fixed-domain comparison class is chosen to force the result.” Deny Definition 1.8 and Remark 1.9; otherwise the object of audit is the least domain in which standing-bearing SM bookkeeping is well-typed.
- “A selector, metadata layer, scheduler, or canonical representative can authorize a further same-scope completion.” Deny Theorem 2.51, Corollary 2.52, or Theorem 2.28; otherwise the device is anchor, tensor, or skin, never a fourth role.
- “A standing failure can be repaired later, localized, factored across modules, or filtered away.” Deny Theorem 2.30; otherwise the proposed repair falls into one of the five closed classes.
- “There is a fifth same-scope locus for new standing-bearing structure.” Deny Theorem 6.8; otherwise the modification moves an anchor, quotient, tensor witness, continuation structure, or has only skin status.
- “A hidden or latent label can survive as same-scope identity-bearing content without appearing in witnesses.” Deny Remark 2.12 and Theorem 6.2; otherwise the label is skin until it changes scope, load, quotient, witness content, or transport/history structure.
- “A new same-domain generator lies outside the listed envelope but still preserves standing.” Deny Theorem 6.2, Remark 6.3, and Corollary 6.4; otherwise the generator factors through normal form, moves a fixed-domain locus, or lacks a transport certificate.
- “A latent non-witnessed label can remain same-scope standing-bearing.” Deny Remark 2.12 and Theorem 1.11; otherwise the label is skin until it becomes a witness by scope or load change.

Counterexample admissibility checklist. A counterexample to the main closure assertion must satisfy all of the following simultaneously: it must keep the declared load fixed; keep the same scope and error envelope; preserve the same witness-output family; preserve transport completeness; preserve the quotient relation $\sim_S$ until a new same-scope witness is supplied; exhibit any new generator as a transport-complete same-domain certificate rather than existence-by-assertion; preserve the admissible continuation/history structure; avoid latent labels that lack invariant witnesses; avoid post-hoc reindexing of committed records; avoid arbitrarily small but real changes in admitted tests as alleged same-scope moves; and avoid selector, typicality, post-selection, or representative-choice

repair. Failure of any item reclassifies the proposal as new load, scope change, skin, or standing failure rather than a counterexample.

A Cubic anomaly check algebra

This appendix records the explicit expansion used in Theorem 4.4 to reduce the cubic anomaly constraint to the quadratic equation (80).

Start from the one-generation anomaly conditions (75)–(78). From (76) and (77) we have

$$Y_L = -3Y_Q, \quad Y_e = -6Y_Q.$$

From (75) we have $Y_d = 2Y_Q - Y_u$. Insert these into (78):

$$0 = 6Y_Q^3 + 2(-3Y_Q)^3 - 3Y_u^3 - 3(2Y_Q - Y_u)^3 - (-6Y_Q)^3 \quad (96)$$

$$= 6Y_Q^3 - 54Y_Q^3 - 3Y_u^3 - 3(8Y_Q^3 - 12Y_Q^2Y_u + 6Y_QY_u^2 - Y_u^3) + 216Y_Q^3 \quad (97)$$

$$= (6 - 54 - 24 + 216)Y_Q^3 + 36Y_Q^2Y_u - 18Y_QY_u^2 \quad (98)$$

$$= 144Y_Q^3 + 36Y_Q^2Y_u - 18Y_QY_u^2. \quad (99)$$

Assuming $Y_Q \neq 0$ and dividing by $-18Y_Q$ gives

$$Y_u^2 - 2Y_QY_u - 8Y_Q^2 = 0,$$

which is (80). Its solutions are $Y_u = 4Y_Q$ or $Y_u = -2Y_Q$, with $Y_d = 2Y_Q - Y_u$.

B Fujikawa construction of the chiral Jacobian and the θ shift

The path-integral measure computation follows the standard Fujikawa method [11, 12].

This appendix gives a line-by-line computation of the Jacobian of the fermionic path-integral measure under a global axial rotation, and the induced shift of the θ term used in Lemma 3.11. We write the calculation in Euclidean signature (where the regulator is well-defined) and then translate the result back to the Minkowski $\theta G\tilde{G}$ normalization.

B.1 Setup: mode expansion and the measure

Consider one Dirac fermion in a background SU(3) gauge field $A_\mu = A_\mu^a T^a$ with T^a Hermitian and $\text{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. Let

$$\not{D} \equiv \gamma^\mu(\partial_\mu - ig_s A_\mu), \quad S_E = \int d^4x \bar{\psi} \not{D} \psi + \dots \quad (100)$$

Choose a complete orthonormal eigenbasis of the Euclidean Dirac operator,

$$\not{D} \phi_n(x) = \lambda_n \phi_n(x), \quad \int d^4x \phi_n^\dagger(x) \phi_m(x) = \delta_{nm}. \quad (101)$$

Expand the Grassmann fields in this basis:

$$\psi(x) = \sum_n a_n \phi_n(x), \quad \bar{\psi}(x) = \sum_n \bar{a}_n \phi_n^\dagger(x), \quad (102)$$

so that (up to an overall constant) the functional measure is

$$\mathcal{D}\psi \mathcal{D}\bar{\psi} = \prod_n da_n d\bar{a}_n. \quad (103)$$

B.2 Axial rotation and the formal Jacobian

Perform a global axial rotation with parameter α ,

$$\psi \mapsto \psi' = e^{i\alpha\gamma_5}\psi, \quad \bar{\psi} \mapsto \bar{\psi}' = \bar{\psi} e^{i\alpha\gamma_5}, \quad (104)$$

(where in Euclidean signature we take $\gamma_5^\dagger = \gamma_5$, $\gamma_5^2 = \mathbf{1}$). At the level of mode coefficients,

$$a'_n = \sum_m \left(\int d^4x \phi_n^\dagger e^{i\alpha\gamma_5} \phi_m \right) a_m \equiv \sum_m (U_\alpha)_{nm} a_m, \quad (105)$$

and similarly $\bar{a}'_n = \sum_m \bar{a}_m (U_\alpha)_{mn}$. Hence

$$\prod_n da'_n = (\det U_\alpha)^{-1} \prod_n da_n, \quad \prod_n d\bar{a}'_n = (\det U_\alpha)^{-1} \prod_n d\bar{a}_n, \quad (106)$$

so the full Jacobian is

$$\mathcal{D}\psi' \mathcal{D}\bar{\psi}' = J(\alpha) \mathcal{D}\psi \mathcal{D}\bar{\psi}, \quad J(\alpha) = (\det U_\alpha)^{-2}. \quad (107)$$

For infinitesimal α , $U_\alpha = \mathbf{1} + i\alpha M + \mathcal{O}(\alpha^2)$ with

$$M_{nm} = \int d^4x \phi_n^\dagger(x) \gamma_5 \phi_m(x), \quad (108)$$

so

$$\ln J(\alpha) = -2 \text{Tr} \ln U_\alpha = -2 \text{Tr}(i\alpha M) + \mathcal{O}(\alpha^2) = -2i\alpha \sum_n \int d^4x \phi_n^\dagger(x) \gamma_5 \phi_n(x) + \mathcal{O}(\alpha^2). \quad (109)$$

The trace $\sum_n \phi_n^\dagger \gamma_5 \phi_n$ is ultraviolet divergent and must be regulated.

B.3 Gauge-covariant regulator and reduction to a local trace

Introduce the gauge-covariant heat-kernel regulator with mass scale M :

$$\sum_n \phi_n^\dagger(x) \gamma_5 \phi_n(x) \mapsto \lim_{M \rightarrow \infty} \sum_n \phi_n^\dagger(x) \gamma_5 e^{-\lambda_n^2/M^2} \phi_n(x) = \lim_{M \rightarrow \infty} \text{Tr} \left[\gamma_5 e^{-\not{p}^2/M^2} \right]_{x,x}. \quad (110)$$

Define the regulated local quantity

$$\mathcal{A}(x) \equiv \lim_{M \rightarrow \infty} \text{Tr} \left[\gamma_5 e^{-\not{p}^2/M^2} \right]_{x,x}. \quad (111)$$

Then, to leading order in α ,

$$\ln J(\alpha) = -2i\alpha \int d^4x \mathcal{A}(x) \quad \Rightarrow \quad J(\alpha) = \exp \left(-2i\alpha \int d^4x \mathcal{A}(x) \right). \quad (112)$$

B.4 Compute $\not{D}^2$ explicitly

We now reduce $\not{D}^2$ to a Laplacian plus a field-strength term. Start from

$$\not{D}^2 = \gamma^\mu \gamma^\nu D_\mu D_\nu, \quad D_\mu = \partial_\mu - ig_s A_\mu. \quad (113)$$

Use the decomposition

$$\gamma^\mu \gamma^\nu = \frac{1}{2} \{\gamma^\mu, \gamma^\nu\} + \frac{1}{2} [\gamma^\mu, \gamma^\nu] = g^{\mu\nu} \mathbf{1} + \sigma^{\mu\nu}, \quad \sigma^{\mu\nu} \equiv \frac{1}{2} [\gamma^\mu, \gamma^\nu], \quad (114)$$

(where in Euclidean signature $g^{\mu\nu} = \delta^{\mu\nu}$; the calculation below is signature-agnostic as long as the standard Clifford algebra holds). Then

$$\not{D}^2 = D^\mu D_\mu + \sigma^{\mu\nu} D_\mu D_\nu. \quad (115)$$

Split $D_\mu D_\nu$ into symmetric and antisymmetric parts:

$$D_\mu D_\nu = \frac{1}{2} \{D_\mu, D_\nu\} + \frac{1}{2} [D_\mu, D_\nu]. \quad (116)$$

Because $\sigma^{\mu\nu}$ is antisymmetric, $\sigma^{\mu\nu} \{D_\mu, D_\nu\} = 0$, so

$$\sigma^{\mu\nu} D_\mu D_\nu = \frac{1}{2} \sigma^{\mu\nu} [D_\mu, D_\nu]. \quad (117)$$

With our convention $D_\mu = \partial_\mu - ig_s A_\mu$, the commutator is

$$[D_\mu, D_\nu] = -ig_s G_{\mu\nu}, \quad G_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu - ig_s [A_\mu, A_\nu] = G_{\mu\nu}^a T^a. \quad (118)$$

Therefore

$$\not{D}^2 = D^2 - \frac{ig_s}{2} \sigma^{\mu\nu} G_{\mu\nu}. \quad (119)$$

B.5 Heat-kernel expansion: isolate the γ_5 term

Insert (119) into (111):

$$\mathcal{A}(x) = \lim_{M \rightarrow \infty} \text{Tr} \left[\gamma_5 \exp \left(-\frac{1}{M^2} \left(D^2 - \frac{ig_s}{2} \sigma^{\mu\nu} G_{\mu\nu} \right) \right) \right]_{x,x}. \quad (120)$$

Expand the exponential in powers of $1/M$ and keep only terms that can contribute to a γ_5 trace. Write

$$\exp \left(-\frac{1}{M^2} (D^2 + X) \right) = e^{-D^2/M^2} \left[\mathbf{1} - \frac{X}{M^2} + \frac{1}{2} \frac{X^2}{M^4} + \mathcal{O} \left(\frac{1}{M^6} \right) \right], \quad X \equiv +\frac{ig_s}{2} \sigma^{\mu\nu} G_{\mu\nu}. \quad (121)$$

Then

$$\begin{aligned} \mathcal{A}(x) &= \lim_{M \rightarrow \infty} \text{Tr} \left[\gamma_5 e^{-D^2/M^2} \right]_{x,x} \\ &\quad - \lim_{M \rightarrow \infty} \frac{1}{M^2} \text{Tr} \left[\gamma_5 e^{-D^2/M^2} X \right]_{x,x} \\ &\quad + \lim_{M \rightarrow \infty} \frac{1}{2M^4} \text{Tr} \left[\gamma_5 e^{-D^2/M^2} X^2 \right]_{x,x} + \dots \end{aligned}$$

The first term vanishes because $\text{Tr}_{\text{spin}}(\gamma_5) = 0$. The second term vanishes because $\text{Tr}_{\text{spin}}(\gamma_5 \sigma^{\mu\nu}) = 0$. Thus the leading contribution comes from the X^2 term:

$$\mathcal{A}(x) = \lim_{M \rightarrow \infty} \frac{1}{2M^4} \text{Tr} \left[\gamma_5 e^{-D^2/M^2} \left(\frac{ig_s}{2} \sigma^{\mu\nu} G_{\mu\nu} \right)^2 \right]_{x,x}. \quad (122)$$

B.6 Coincident-point kernel and the momentum integral

To extract the leading $M \rightarrow \infty$ limit, use that the coincident-point kernel of e^{-D^2/M^2} is dominated by short distances. To the order needed in (122), one may replace D^2 by the free Laplacian in the coincident kernel, because any insertion of gauge fields from D^2 brings additional powers of $1/M$ that vanish in the $M \rightarrow \infty$ limit. Thus,

$$\langle x | e^{-D^2/M^2} | x \rangle = \int \frac{d^4 k}{(2\pi)^4} e^{-k^2/M^2} + \mathcal{O}\left(\frac{1}{M^2}\right) = \frac{M^4}{16\pi^2} + \mathcal{O}(M^2). \quad (123)$$

(We used $\int d^4 k e^{-k^2/M^2} = \pi^2 M^4$ and divided by $(2\pi)^4$.)

Insert (123) into (122) and pull out the slowly varying fields:

$$\mathcal{A}(x) = \lim_{M \rightarrow \infty} \frac{1}{2M^4} \left(\frac{M^4}{16\pi^2} \right) \text{Tr}_{\text{spin,color}} \left[\gamma_5 \left(\frac{ig_s}{2} \sigma^{\mu\nu} G_{\mu\nu} \right)^2 \right] + \mathcal{O}\left(\frac{1}{M^2}\right). \quad (124)$$

B.7 Spinor and color traces

Compute the trace in (124). First expand

$$\left(\frac{ig_s}{2} \sigma^{\mu\nu} G_{\mu\nu} \right)^2 = -\frac{g_s^2}{4} \sigma^{\mu\nu} \sigma^{\rho\sigma} G_{\mu\nu}^a G_{\rho\sigma}^b T^a T^b. \quad (125)$$

Therefore

$$\text{Tr}_{\text{spin,color}} \left[\gamma_5 \left(\frac{ig_s}{2} \sigma^{\mu\nu} G_{\mu\nu} \right)^2 \right] = -\frac{g_s^2}{4} \text{Tr}_{\text{spin}}(\gamma_5 \sigma^{\mu\nu} \sigma^{\rho\sigma}) \text{Tr}_{\text{color}}(T^a T^b) G_{\mu\nu}^a G_{\rho\sigma}^b. \quad (126)$$

Using $\text{Tr}_{\text{color}}(T^a T^b) = \frac{1}{2} \delta^{ab}$, this becomes

$$\text{Tr}_{\text{spin,color}}[\dots] = -\frac{g_s^2}{8} \text{Tr}_{\text{spin}}(\gamma_5 \sigma^{\mu\nu} \sigma^{\rho\sigma}) G_{\mu\nu}^a G_{\rho\sigma}^a. \quad (127)$$

Now compute the spinor trace. Using $\sigma^{\mu\nu} = \frac{1}{2}[\gamma^\mu, \gamma^\nu]$ and cyclicity,

$$\begin{aligned} \text{Tr}_{\text{spin}}(\gamma_5 \sigma^{\mu\nu} \sigma^{\rho\sigma}) &= \frac{1}{4} \text{Tr}_{\text{spin}}(\gamma_5 [\gamma^\mu, \gamma^\nu] [\gamma^\rho, \gamma^\sigma]) \\ &= \frac{1}{4} \text{Tr}_{\text{spin}}(\gamma_5 (\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) (\gamma^\rho \gamma^\sigma - \gamma^\sigma \gamma^\rho)) \\ &= \frac{1}{4} \left(\text{Tr}(\gamma_5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) - \text{Tr}(\gamma_5 \gamma^\mu \gamma^\nu \gamma^\sigma \gamma^\rho) \right. \\ &\quad \left. - \text{Tr}(\gamma_5 \gamma^\nu \gamma^\mu \gamma^\rho \gamma^\sigma) + \text{Tr}(\gamma_5 \gamma^\nu \gamma^\mu \gamma^\sigma \gamma^\rho) \right). \end{aligned} \quad (128)$$

Use the standard identity (applicable in four dimensions with $\epsilon^{0123} = +1$ in Minkowski signature, and with corresponding Euclidean continuation)

$$\text{Tr}(\gamma_5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4 \epsilon^{\mu\nu\rho\sigma}. \quad (129)$$

Because $\epsilon^{\mu\nu\sigma\rho} = -\epsilon^{\mu\nu\rho\sigma}$ and $\epsilon^{\nu\mu\rho\sigma} = -\epsilon^{\mu\nu\rho\sigma}$, each of the four terms contributes the same $\epsilon^{\mu\nu\rho\sigma}$. Therefore

$$\text{Tr}_{\text{spin}}(\gamma_5 \sigma^{\mu\nu} \sigma^{\rho\sigma}) = 4 \epsilon^{\mu\nu\rho\sigma}. \quad (130)$$

(If one uses a different γ_5 convention, the overall sign is absorbed into the definition of $\tilde{G}^{\mu\nu}$ and the sign of θ ; the physical invariant $\bar{\theta}$ in Lemma 3.11 is unaffected.)

Insert (130) into (127):

$$\text{Tr}_{\text{spin,color}}[\cdots] = -\frac{g_s^2}{8} (4\epsilon^{\mu\nu\rho\sigma}) G_{\mu\nu}^a G_{\rho\sigma}^a = -\frac{g_s^2}{2} \epsilon^{\mu\nu\rho\sigma} G_{\mu\nu}^a G_{\rho\sigma}^a. \quad (131)$$

Using $\tilde{G}^{a\mu\nu} \equiv \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}^a$, we have $\epsilon^{\mu\nu\rho\sigma} G_{\mu\nu}^a G_{\rho\sigma}^a = 2 G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$. Thus

$$\text{Tr}_{\text{spin,color}}[\cdots] = -g_s^2 G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (132)$$

B.8 Assemble the coefficient

Combine (124) and (132):

$$\mathcal{A}(x) = \frac{1}{2} \cdot \frac{1}{16\pi^2} \cdot \left(-g_s^2 G_{\mu\nu}^a \tilde{G}^{a\mu\nu} \right) = -\frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (133)$$

For N_f flavors rotated simultaneously, $\mathcal{A}(x)$ is multiplied by N_f .

Insert this into (112):

$$J(\alpha) = \exp\left(+2i\alpha N_f \int d^4x \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}\right), \quad (134)$$

where the sign follows our conventions above.

B.9 Read off the θ shift

The Jacobian contributes to the effective action as $\Delta S_E = -\ln J(\alpha)$, i.e.

$$\Delta S_E = -2i\alpha N_f \int d^4x \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (135)$$

Translating back to Minkowski signature and comparing with

$$\mathcal{L}_\theta = \frac{g_s^2}{32\pi^2} \theta G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad (136)$$

we obtain the shift

$$\theta \mapsto \theta + 2N_f \alpha, \quad (137)$$

which is the step used in Lemma 3.11.

C Broken-gauge Ward/Slavnov–Taylor identity and longitudinal–Goldstone identity transport

The identity-transport theorem (external bridge language: equivalence theorem) is standard; see e.g. [13, 14].

This appendix gives a self-contained construction sequence of the identity used in Lemma 3.13. We work in an R_ξ gauge and use BRST invariance to derive a Slavnov–Taylor relation that links amplitudes with an external longitudinal massive gauge boson to amplitudes with the corresponding would-be Goldstone boson. We then take the high-energy limit $E \gg m_W$ to obtain the identity transport of longitudinal and Goldstone amplitude classes.

C.1 Gauge fixing with an auxiliary field

Consider (for definiteness) the $SU(2)$ part of the electroweak sector after spontaneous symmetry breaking, with massive gauge bosons W_μ^a and would-be Goldstones π^a (the construction sequence applies componentwise to $W^\pm$ and Z). Choose the R_ξ gauge-fixing function

$$F^a \equiv \partial^\mu W_\mu^a + \xi m_W \pi^a. \quad (138)$$

Introduce the Nakanishi–Lautrup auxiliary field B^a and write the gauge-fixing Lagrangian as

$$\mathcal{L}_{\text{gf}} = B^a F^a + \frac{\xi}{2} B^a B^a. \quad (139)$$

Eliminating B^a by its algebraic equation of motion gives $B^a = -\frac{1}{\xi} F^a$ and hence

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} (F^a)^2 = -\frac{1}{2\xi} (\partial \cdot W^a + \xi m_W \pi^a)^2, \quad (140)$$

which is the standard R_ξ gauge fixing used in the main text.

C.2 Ghost term and BRST transformations

Introduce Faddeev–Popov ghosts c^a and antighosts $\bar{c}^a$. The gauge-fixed action is made BRST invariant by adding the ghost term

$$\mathcal{L}_{\text{gh}} = -\bar{c}^a \frac{\delta F^a}{\delta \theta^b} c^b, \quad (141)$$

where θ^b are the infinitesimal gauge parameters and $\delta F^a / \delta \theta^b$ is computed from the gauge transformations of W_μ^a and π^a .

BRST transformations s are defined so that gauge transformations are reproduced with ghost parameters and $s^2 = 0$. The only pieces we will need explicitly are the linearized transformations around the vacuum (the nonlinear terms are fixed by the gauge algebra but do not enter the identity we derive):

$$s W_\mu^a = \partial_\mu c^a + \dots, \quad (142)$$

$$s \pi^a = -m_W c^a + \dots, \quad (143)$$

$$s c^a = -\frac{g_2}{2} \epsilon^{abc} c^b c^c, \quad (144)$$

$$s \bar{c}^a = B^a, \quad s B^a = 0. \quad (145)$$

A key structural fact is that the gauge-fixing plus ghost Lagrangian is BRST-exact:

$$\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{gh}} = s \left[\bar{c}^a \left(F^a + \frac{\xi}{2} B^a \right) \right]. \quad (146)$$

BRST invariance of the full gauge-fixed action follows immediately from $s^2 = 0$.

C.3 F^a is BRST-exact on shell

Because $s \bar{c}^a = B^a$ and B^a is algebraic, the B equation of motion gives

$$B^a = -\frac{1}{\xi} F^a. \quad (147)$$

Therefore

$$F^a = -\xi B^a = -\xi s \bar{c}^a. \quad (148)$$

Thus F^a is BRST-exact (after eliminating the auxiliary field). This is the algebraic origin of the Slavnov–Taylor identity relating $\partial \cdot W^a$ and π^a .

C.4 Vanishing of BRST-exact insertions between physical states

Physical asymptotic states are taken to be BRST-closed (they are annihilated by the BRST charge) and BRST-exact states are regarded as null. Consequently, matrix elements of BRST-exact operators between physical states vanish:

$$\langle \text{phys} | s(\mathcal{O}) | \text{phys} \rangle = 0. \quad (149)$$

Applying (149) to $\mathcal{O} = \bar{c}^a$ and using (148) yields

$$\langle \text{phys} | F^a(x) | \text{phys} \rangle = 0 \quad \Rightarrow \quad \langle \text{phys} | \partial^\mu W_\mu^a(x) | \text{phys} \rangle = -\xi m_W \langle \text{phys} | \pi^a(x) | \text{phys} \rangle. \quad (150)$$

Equation (150) is the configuration-space form of the Ward/Slavnov–Taylor identity needed.

C.5 From $\partial \cdot W^a$ insertions to amputated amplitudes

Let $\mathcal{G}_\mu^a(x; \{y\})$ denote a connected Green function with one $W_\mu^a(x)$ insertion and other insertions collectively denoted by $\{y\}$ (all taken to correspond to physical external legs in an S -matrix element). Fourier transforming the x dependence gives

$$\tilde{\mathcal{G}}_\mu^a(p; \{y\}) = \int d^4x e^{ip \cdot x} \mathcal{G}_\mu^a(x; \{y\}). \quad (151)$$

Then the Fourier transform of $\partial^\mu W_\mu^a$ corresponds to multiplication by ip^μ :

$$\tilde{\mathcal{G}}_{\partial \cdot W}^a(p; \{y\}) \equiv \int d^4x e^{ip \cdot x} \langle 0 | T \{ \partial^\mu W_\mu^a(x) \cdots \} | 0 \rangle = ip^\mu \tilde{\mathcal{G}}_\mu^a(p; \{y\}). \quad (152)$$

Likewise define $\tilde{\mathcal{G}}_\pi^a(p; \{y\})$ with a π^a insertion.

Equation (150) implies, at the level of such Green functions with only physical external legs,

$$ip^\mu \tilde{\mathcal{G}}_\mu^a(p; \{y\}) = -\xi m_W \tilde{\mathcal{G}}_\pi^a(p; \{y\}). \quad (153)$$

Now apply LSZ reduction. For an external massive vector boson, the amputated object that becomes the scattering amplitude is obtained by multiplying by $(p^2 - m_W^2)$ and taking the on-shell limit; for an external scalar Goldstone, one multiplies by $(p^2 - \xi m_W^2)$ (because in R_ξ gauge the Goldstone propagator pole is at $p^2 = \xi m_W^2$). Denote the amputated, on-shell objects by $\mathcal{M}_\mu(W^a(p), \dots)$ and $\mathcal{M}(\pi^a(p), \dots)$. The amputated version of (153) becomes the standard Slavnov–Taylor relation

$$p^\mu \mathcal{M}_\mu(W^a(p), \dots) = i m_W \mathcal{M}(\pi^a(p), \dots), \quad (154)$$

where we set $\xi = 1$ in the final on-shell relation so that the Goldstone pole coincides with m_W ('t Hooft–Feynman gauge). More general ξ versions differ by ξ -dependent normalization factors that cancel from physical S -matrix elements; the high-energy identity-transport statement below is unchanged in its standing-relevant content.

C.6 Longitudinal polarization and the high-energy limit

For an on-shell massive vector with momentum $p^\mu = (E, \mathbf{p})$ and $p^2 = m_W^2$, a longitudinal polarization can be chosen as

$$\epsilon_L^\mu(p) = \frac{1}{m_W} (|\mathbf{p}|, E \hat{\mathbf{p}}), \quad \hat{\mathbf{p}} = \mathbf{p}/|\mathbf{p}|. \quad (155)$$

One checks directly that $p \cdot \epsilon_L = 0$ and $\epsilon_L^2 = -1$ (Minkowski signature). In the high-energy regime $E \gg m_W$, we have $E = |\mathbf{p}| + \mathcal{O}(m_W^2/E)$, so

$$\epsilon_L^\mu(p) = \frac{p^\mu}{m_W} + \mathcal{O}\left(\frac{m_W}{E}\right). \quad (156)$$

C.7 Identity-transport theorem (tree level, audit form)

Combine (154) with (156):

$$\begin{aligned}
\mathcal{M}(W_L^a(p), \dots) &= \epsilon_L^\mu(p) \mathcal{M}_\mu(W^a(p), \dots) \\
&= \left(\frac{p^\mu}{m_W} + \mathcal{O}\left(\frac{m_W}{E}\right) \right) \mathcal{M}_\mu(W^a(p), \dots) \\
&= \frac{1}{m_W} (im_W \mathcal{M}(\pi^a(p), \dots)) + \mathcal{O}\left(\frac{m_W}{E}\right) \\
&= i \mathcal{M}(\pi^a(p), \dots) + \mathcal{O}\left(\frac{m_W}{E}\right).
\end{aligned} \tag{157}$$

The overall phase i depends only on the convention used to define the scalar vs. vector LSZ normalizations; at tree level and with consistent conventions it may be absorbed so that the standing-relevant statement is:

$$\mathcal{M}(W_L^a(p), \dots) = \mathcal{M}(\pi^a(p), \dots) + \mathcal{O}\left(\frac{m_W}{E}\right). \tag{158}$$

Equation (158) is precisely the amplitude-class identification used in Lemma 3.13.

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